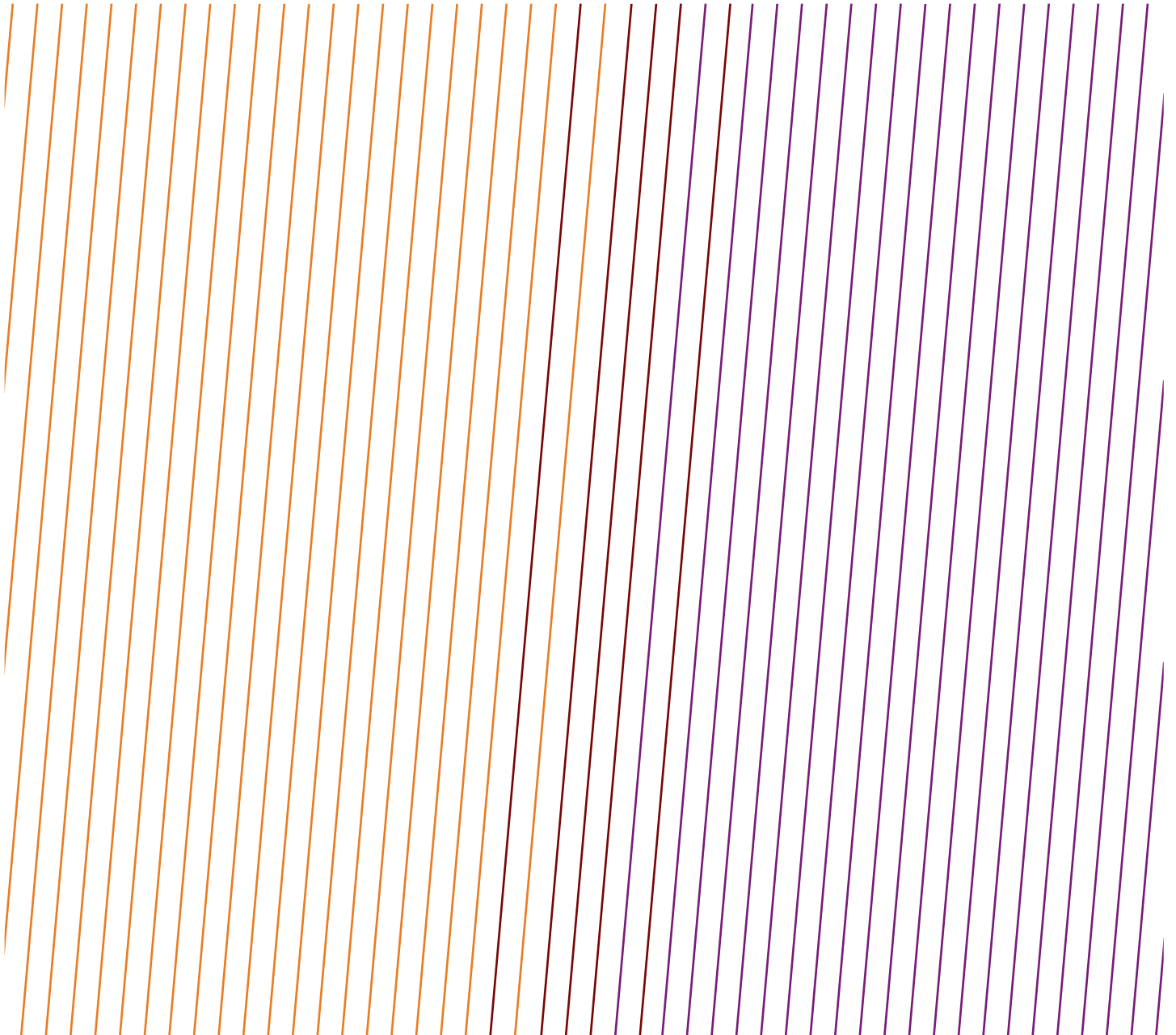


DATA SERIES

Safety performance indicators – Fatality and permanent impairment injuries (FPI) – 2024 data



Acknowledgements

IOGP is grateful for the participation of the companies that have submitted safety performance indicators. This Report was produced by the Safety Committee.

Feedback

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DATA SERIES

Safety performance indicators – Fatality and permanent impairment injuries (FPI) – 2024 data

Revision history

VERSION	DATE	AMENDMENTS
1.0	December 2025	First release

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Contributing companies

The fatality and permanent impairment (FPI) statistics in this Report for 2024 were derived from voluntary data submissions to IOGP by the following companies:

ADDAX Petroleum Limited	Dolphin Energy	Petrobras
ADNOC	ENI	Pluspetrol
Aker BP	Equinor ASA	Prime Energy
Apache	ExxonMobil	Prio
Assala Energy	Gulf Keystone	PTTEP
Azule	Harbour Energy	QatarEnergy LNG
Bapco Energies	Hess Corporation	Repsol
Basrah Gas Company	INPEX Corporation	Shell Companies
Beach Energy	Kosmos Energy	SOCAR
bp	Kuwait Oil Company	Sonangol
CCED	MOL	Spirit Energy
Cenovus	North Oil Company	TotalEnergies
Chevron	ONGC	Trident Energy
CNOOC	Orlen	Tullow Oil
ConocoPhillips	Oxy	Vår Energy
Crescent Petroleum	Pan American Energy	Woodside
Dana Gas	Perenco	YPF SA

Introduction

IOGP has set a course for our industry to focus on preventing life-changing injuries and fatalities. In 2022, an Expert Group was set up to agree upon a common industry definition of actual, potential, and near miss fatality and permanent impairment (FPI) injuries.

Although significant progress has been made by IOGP Member Companies in reducing the lost time injury rate (LTIR) and total recordable injury rate (TRIR), we have reached a plateau, as shown in Figure 1.

While a reduction in overall injury rates is positive, not all recordable or even lost time injuries are equivalent. For the purposes of TRIR, an injury that results in a single stitch to a finger has the same statistical weighting in the rate calculation as a fatality. A rate with such a broad scope of injury classification does not give any indication of injury severity. Addressing the most severe injuries, those that change people's lives, needed to be prioritized, i.e., focus on what matters most.

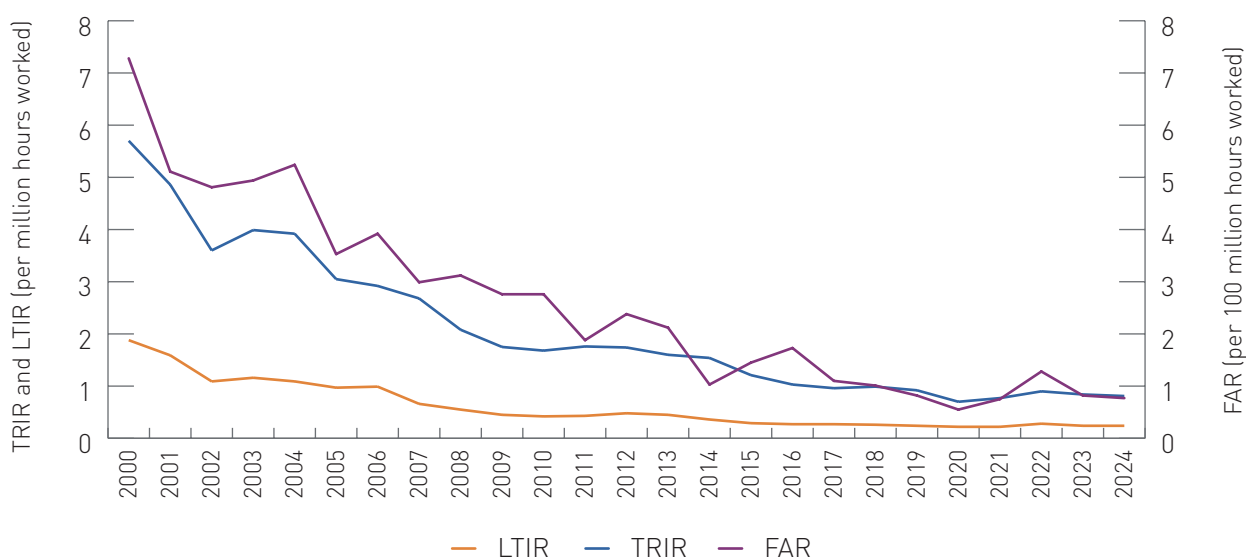


Figure 1: FAR, TRIR and LTIR, 2000-2024

Focusing on controls and safeguards to primarily prevent fatalities has been shown to have reduced the number of recordable injuries¹, but many organizations commit much of their Health, Safety, and Environment (HSE) resources trying to reduce the number of recordable injuries. This focus on reducing high-volume, lower severity injuries (as shown on the right-hand side of Figure 2), such as slips and trips (at the same height), cuts, punctures and scrapes, overexertion, strain, etc., may have the desired effect on reducing TRIR, but it is highly unlikely to reduce the number of life-changing injuries and fatalities.

¹ Peuscher, W., Groeneweg J. 2012.

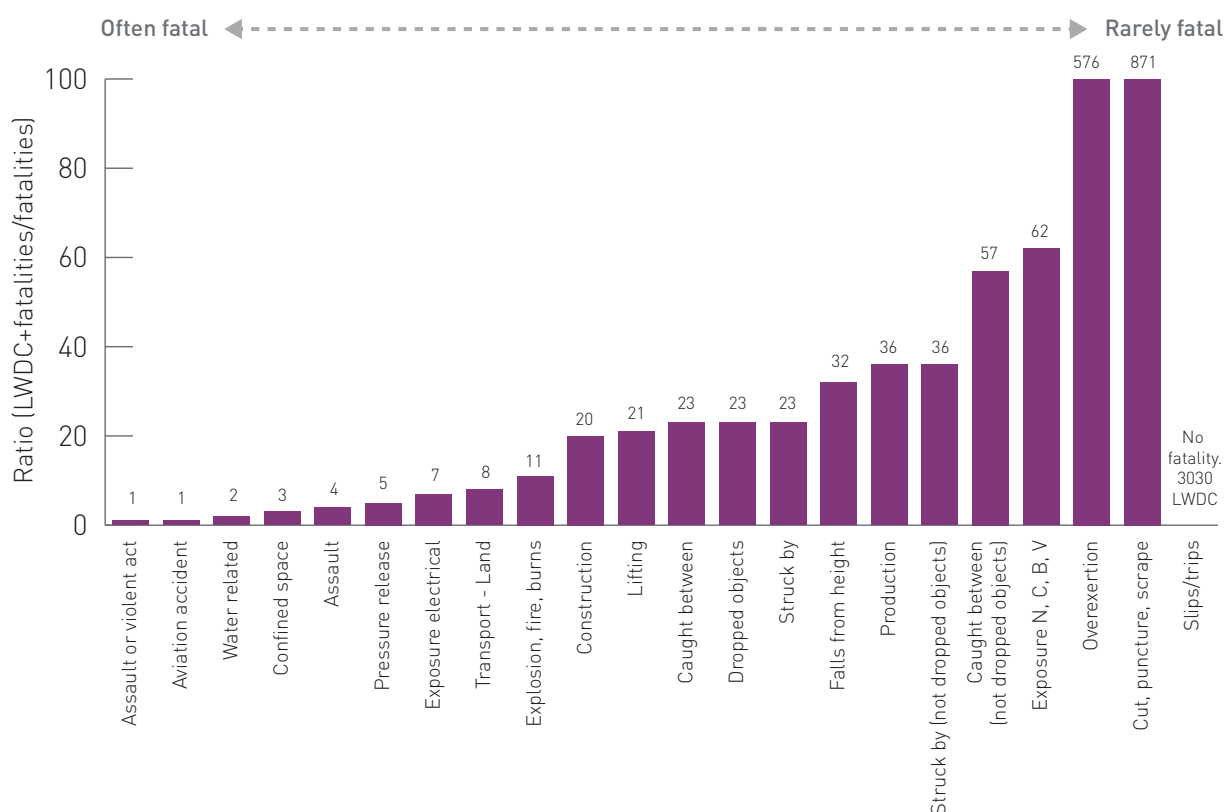


Figure 2: Fatality to LTI (fatality + lost work day case) ratio by cause or activity, 2009 - 2024

The IOGP FPI definitions are designed for use in energy industry reporting. They are deliberately concise and concept based. The concepts are provided in plain language to facilitate reliable interpretation and application across a wide range of potential scenarios by industry personnel without health qualifications. The classification of a permanent impairment injury as per the definition should be determinable from a basic assessment of known injuries assisted by injury reports, medical certificates, and/or worker self-reports of current status.

These definitions are aligned to recognized international systems of classification including:

- American Medical Association (AMA) Guides to the Evaluation of Permanent Impairment²
- World Health Organization (WHO) International Classification of Functioning, Disability and Health (ICF)

The scope of this Report includes three years of data which is insufficient for meaningful trend analysis but does demonstrate the consistent classification and understanding by reporting companies of the new PI injury definition.

Quality Assurance (QA)/Quality Control (QC) was carried out on the data submitted to assure consistency in classification. The definition used for this Report is in Section 1.

For additional background on the development of the FPI definition and data collection, see Walker et al, (2024).

Countries and companies that are subject to international sanctions (Russia and Iran) are not featured in this Report.

² AMA, 6th Edition.

1. FPI definitions and criteria

1.1 Definition

The definition for FPI injury data submitted by IOGP Member Companies in this Report is below.

Fatality

A death as a result of a work-related incident. 'Delayed' deaths that occur within 180 days post incident are to be included if the death was a direct result of the incident (refer to the annual IOGP Safety Performance Indicators User Guide for the definition and further guidance on fatality, 'delayed' deaths, work-related, and injury).

Permanent impairment (PI)

Permanent impairment (PI) is defined as the outcome of a work-related injury from which the worker cannot or is not expected to return to their previous (pre-incident) whole person function as a result of an acute, single incident, resulting in any of the following:

- permanent loss of body parts
- permanent reduction of organ's physiological function
- permanent reduction in skin and musculoskeletal function
- permanent reduction in psychological, social, or cognitive function

For reporting purposes, the classification is to be made within 180 days of the injury.

"Whole person function" relates to a person's physiological/psychological status.

Impairment is defined as either, or both:

- loss of ability to return to the previous work role
- loss of ability to perform the activities of daily living (ADL)

1.2 Permanent impairment criteria

The permanent impairment metric is met by any of the below criteria singularly or in combination. Impairment may occur to physical, psychological, or social function if related to an acute/single incident.

Permanent loss of body parts

Permanent loss of body parts as a result of an acute/single incident due to mechanical trauma, thermal trauma, radioactivity, vibration, chemical exposure, or other, including but not limited to:

- amputations with loss of bone, inclusive of one or more joints (not able to be reattached)
- loss of eye(s)

- loss of internal organs, e.g., kidney, spleen (may also meet criteria for permanent reduction of organ function)
- loss of facial structures, e.g., nose, ears, leading to disfigurement
- loss of genitalia or other sexual organs
- successful use of prostheses (for functional or cosmetic purposes) does not negate a permanent impairment definition

If any of the above are met, the injury is defined as permanent impairment. For any other injury resulting in loss of body parts, when classification of permanent impairment is unclear, a medical professional's opinion should be used to determine permanent impairment.

Permanent reduction of organ function

Permanent reduction in an organ's physiological function as a result of an acute/single incident due to a mechanical trauma, thermal trauma, radioactivity, vibration, chemical exposure or other, including but not limited to:

- eye injury with permanent loss of vision
- bowel injury requiring permanent stoma
- spinal injury requiring ongoing use of wheelchair
- brain injury causing ongoing epilepsy (recurrent seizures)
- heart injury with impaired exertion tolerance/ measurable permanent loss of function
- lung injury, including inhalation, with impaired exertion tolerance/measurable permanent loss of function
- injury causing loss of smell that creates an ongoing safety concern or significant personal impact
- ear injury with permanent loss of hearing (at least equivalent to an OSHA recordable shift), or persistent tinnitus (ear ringing) post injury after exposure to a blast
- post injury complex pain disorders with ongoing uncontrolled pain and/or disrupted nerve and musculoskeletal function
- genital or other sexual organ injury leading to permanent loss of function

If any of the above are met, the injury is defined as permanent impairment. For any other injury resulting in permanent reduction of organ function, when classification of permanent impairment is unclear, a medical professional's opinion should be used to determine permanent impairment.

Permanent reduction in skin and/or musculoskeletal function

Permanent reduction in skin and/or musculoskeletal function as a result of an acute/single incident due to a mechanical trauma, thermal trauma, radioactivity, vibration, chemical exposure, or other, including but not limited to:

- Deep burns with scarring or skin grafts that permanently impair body movement or other function. These are usually 2nd or 3rd degree burns and may be caused by thermal or chemical exposure. This includes a reduction of skin integrity, hypersensitivity to touch, or skin fragility leading to friction ulceration. Burns that cause significant disfigurement or change to physical appearance resulting in a permanent impairment due to psychological, social, or cognitive function would be considered under these criteria.
- Tendon/ligament tear with ongoing abnormal joint range of movement for normal work or general function, e.g., cannot reach overhead, cannot squat, or cannot kneel.
- Traumatic soft tissue injury to shoulder/knee/elbow with ongoing abnormal joint range of movement.
- Fracture with ongoing impairment to mobility.
- Amputation with successful reattachment but permanent abnormal function.

If any of the above are met, the injury is defined as permanent impairment. For any other injury resulting in permanent reduction in skin and/or musculoskeletal function, when classification of permanent impairment is unclear, a medical professional's opinion should be used to determine permanent impairment.

Permanent reduction in psychological, social, or cognitive function

Permanent reduction in psychological, social, or cognitive function as a result of an acute/single incident with psychological trauma, including an inability to self-care, and/or an inability to maintain previous work, social, and community relationships, post incident, including but not limited to:

- brain injury with ongoing impact to emotional regulation, speech, memory, or other cognitive dysfunction
- post injury psychiatric disorders including Post Traumatic Stress Disorder (PTSD) with inability to return to previous work role and/or schedule

If any of the above are met, the injury is defined as permanent impairment. For any other injury resulting in permanent reduction in psychological, social, or cognitive function, when classification of permanent impairment is unclear, a medical professional's opinion should be used to determine permanent impairment.

Examples, associated Frequently Asked Questions (FAQ) that have helped support Member Companies in the classification process, and any future updates to the definition, are available at <https://iogp.org/fpi>.

1.3 Rate calculations

Actual FPI metrics can also be shown as a rate. IOGP has developed fatality and permanent impairment rate (FPIR) and permanent impairment rate (PIR) metrics using the same approach as fatal accident rate (FAR). Rates use 100 million work hours as the denominator, as the number of actual PI injuries reported is of the same order as fatalities.

Rate calculations

Permanent impairment rate (PIR)

Number of company/contractor PI injuries per 100 million hours worked

Fatal accident rate (FAR)

The number of company/contractor fatalities* per 100 million hours worked

Fatality and permanent impairment rate (FPIR)

Number of company/contractor fatalities* plus PI injuries per 100 million hours

*In this Report, excludes fatalities where PI data (zero or greater) were not reported.

2. Scope of data submissions

A summary of the scope of the 2022-2024 data submissions is shown in Tables 1 and 2.

Not all companies that reported safety performance data have participated in the voluntary reporting of permanent impairment data. The information shown in this Report is therefore a subset of the data in the annual *Safety Performance Indicators* Report. The specific mid-year FPI data collection also offers reporting companies the opportunity to capture any delayed deaths from the previous year. Therefore, the two data sets should not be directly compared.

Table 1: 2022 - 2024 Scope of IOGP Annual Safety Performance indicator data

IOGP annual safety performance indicator data (Reports 2022s, 2023s, and 2024s)	2022	2023	2024
Total number of IOGP Member Companies (eligible to report, i.e., excludes associate and association members) at the time of data collection	65	71	72
Number of Member Companies reporting	51	58	56
Work hours (millions)	2,579	3,291	4,159
Countries	92	92	87
Fatalities	33	27	32
FAR – fatalities per 100 million hours	1.28	0.82	0.77

The scope of data used throughout the analysis within this Report is shown in Table 2.

Table 2: 2022 - 2024 Scope of IOGP Member Company FPI data

Scope of FPI data used in this report (excludes companies not reporting permanent impairment data)	2022	2023	2024
Number of Member Companies reporting	37	57	51
Work hours (millions)	2,331	3,254	3,417
Percentage of total hours in the safety performance indicator database (Table 1)	90%	99%	82%
Countries	90	90	84
Fatalities	29	27	27
PI injuries reported	16	31	35
PIR – PI injuries per 100 million hours	0.69	0.95	1.02
FPIR – FPI per 100 million hours	1.93	1.78	1.81
FAR – fatalities per 100 million hours	1.24	0.83	0.79

Countries and companies that are subject to international sanctions (Russia and Iran) are not featured in this Report.

3. 2024 results - overall

3.1 Number of fatalities and permanent impairment injuries

Table 3 shows the total number of fatalities and PI injuries, and the number of incidents in which these occurred. The number of FPI injuries will always be equal to, or larger than, the number of incidents

Table 3: Number of fatalities and permanent impairment injuries

Company/ contractor	2022			2023			2024		
	Onshore	Offshore	Overall	Onshore	Offshore	Overall	Onshore	Offshore	Overall
Fatalities									
Company	2	0	2	4	2	6	3	2	5
Contractor	20	7	27	18	3	21	12	10	22
OVERALL fatalities	22	7	29	22	5	27	15	12	27
PI injuries									
Company	0	1	1	5	0	5	3	2	5
Contractor	14	1	15	13	13	26	24	6	30
OVERALL PI	14	2	16	18	13	31	27	8	35
Fatalities and PI injuries									
Company	2	1	3	9	2	11	6	4	10
Contractor	34	8	42	31	16	47	36	16	52
OVERALL	36	9	45	40	18	58	42	20	62

Table 4: Number of fatal or permanent impairment incidents

FPI incidents*									
Company	1	1	2	8	1	9	6	3	9
Contractor	31	7	38	23	16	39	32	10	42
OVERALL FPI incidents	32	8	40	31	17	48	38	13	51

* More than one FPI injury may be a result of a single incident.

3.2 Fatal accident rate (FAR)

Rate calculations

Fatal accident rate (FAR)

The number of company/contractor fatalities per 100 million hours worked

The annual FAR for the FPI data reported, as per Table 2 in Section 2 Scope, is shown in Table 5.

Table 5: Fatal accident rate

Fatal accident rate (FAR)			
	2022	2023	2024
Company	0.34	0.69	0.65
Contractor	1.55	0.88	0.83
OVERALL	1.24	0.83	0.79
Onshore	1.29	0.92	0.58
Offshore	1.11	0.59	1.45

3.3 Permanent impairment rate (PIR)

Rate calculations

Permanent impairment rate (PIR)

Number of company/contractor PI injuries per 100 million hours worked

The annual PIR for the scope of FPI data reported, as per Table 2 in Section 2 Scope, is shown in Table 6.

Table 6: Permanent impairment rate

Permanent impairment rate (PIR)			
	2022	2023	2024
Company	0.17	0.58	0.65
Contractor	0.86	1.09	1.13
OVERALL	0.69	0.95	1.02
Onshore	0.82	0.75	1.04
Offshore	0.32	1.53	0.97

3.4 Fatality and permanent impairment rate (FPIR)

Rate calculations

Fatality and permanent impairment rate (FPIR)

Number of company/contractor fatalities plus PI injuries per 100 million hours

The annual FPIR for the scope of FPI data reported, as per Table 2 in Section 2 Scope of data submissions, is shown in Table 7.

Table 7: Fatality and permanent impairment rate

Fatality and permanent impairment rate (FPIR)			
	2022	2023	2024
Company	0.51	1.27	1.31
Contractor	2.41	1.97	1.96
OVERALL	1.93	1.78	1.81
Onshore	2.12	1.66	1.62
Offshore	1.43	2.12	2.42

3.5 Fatalities and PI injuries by cause and activity

Table 8 shows the number of fatalities and PI injuries by cause and activity, aggregated for all years of reporting. Data are shown separately for fatality and PI injuries for each year of reporting in Appendix B, Tables B4 and B5.

Table 8: Number of fatalities and permanent impairment injuries by cause and activity 2022-2024

Activity	Cause																
	Assault or violent act	Aviation accident	Caught in, under or between (excl. dropped objects)	Confined space	Cut, puncture, scrape	Dropped objects	Explosion, fire or burns	Exposure electrical	Exposure noise, chemical, biological, vibration, extreme temperature	Falls from height	Overexertion, strain	Pressure release	Slips and trips (at same height)	Struck by (not dropped object)	Water related, drowning	Unspecified - Other	Total
Construction, commissioning, decommissioning	4		6			2		1		1	1	4		2			21
Drilling, workover, well operations			18		1	4	9		2	5		1	1	7			48
Excavation, trenching, ground disturbance			1														1
Lifting, crane, rigging, deck operations			5			6		1						2	2		16
Maintenance, inspection, testing			6	1		3	3	3	2	3	1	2		3			27
Office, warehouse, accommodation, catering					1												1
Production operations	7			1			7					2	1	3			21
Transport - Air		3													1		4
Transport - Land	2		6			1	2							2	1		14
Transport - Water, incl. marine activity			5				1							3			9
Unspecified - other			1											1		1	3
Total	13	3	48	2	2	16	22	5	4	9	2	9	2	23	4	1	165

By cause, in 2024, 54% of the reported PI were categorized as 'caught in, under or between (excluding dropped objects)' (42% in 2023, 32% in 2022), and 11% were categorized as 'explosion, fire or burns' (3% in 2023, 0 in 2022). Dropped objects and Struck by (not dropped object) each accounted for 9% of the PI reported in 2023 (7% and 26% in 2023, 19% and 6% in 2022).

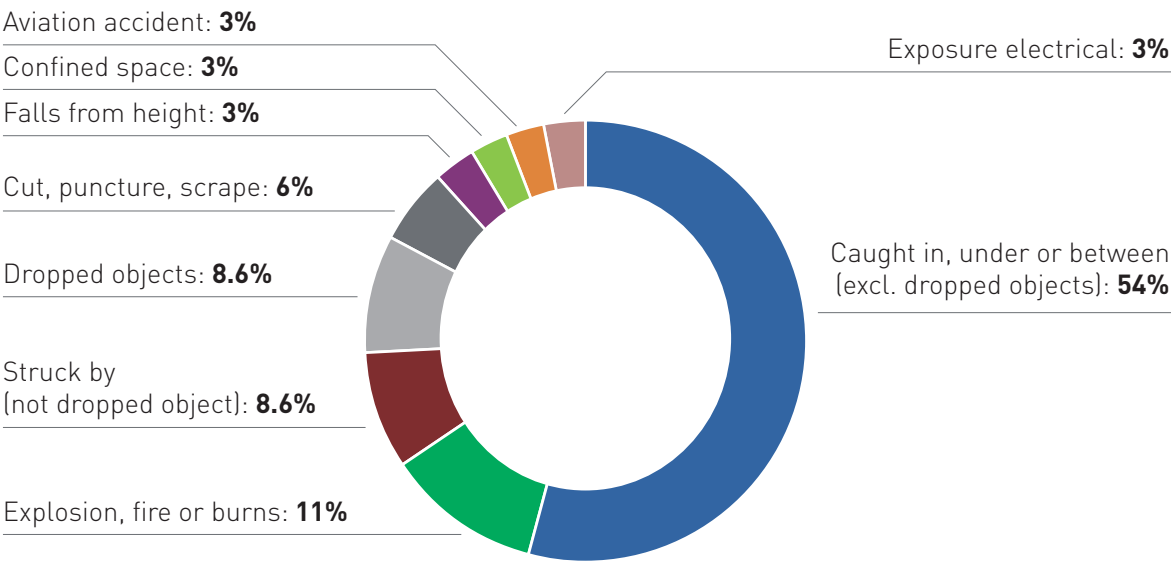


Figure 3: PI injuries by cause 2024

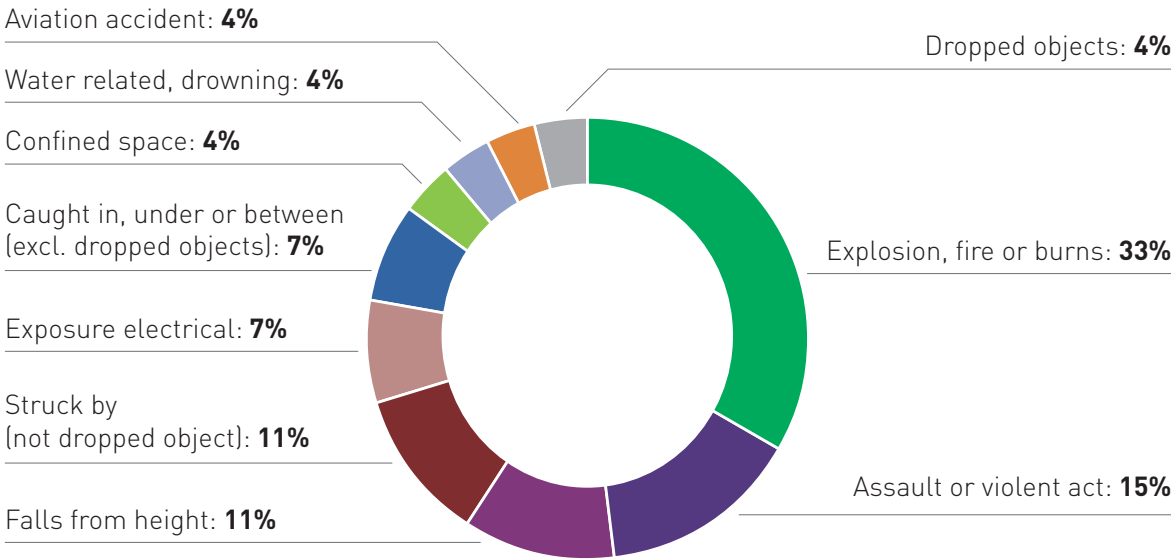


Figure 4: Fatalities by cause 2024

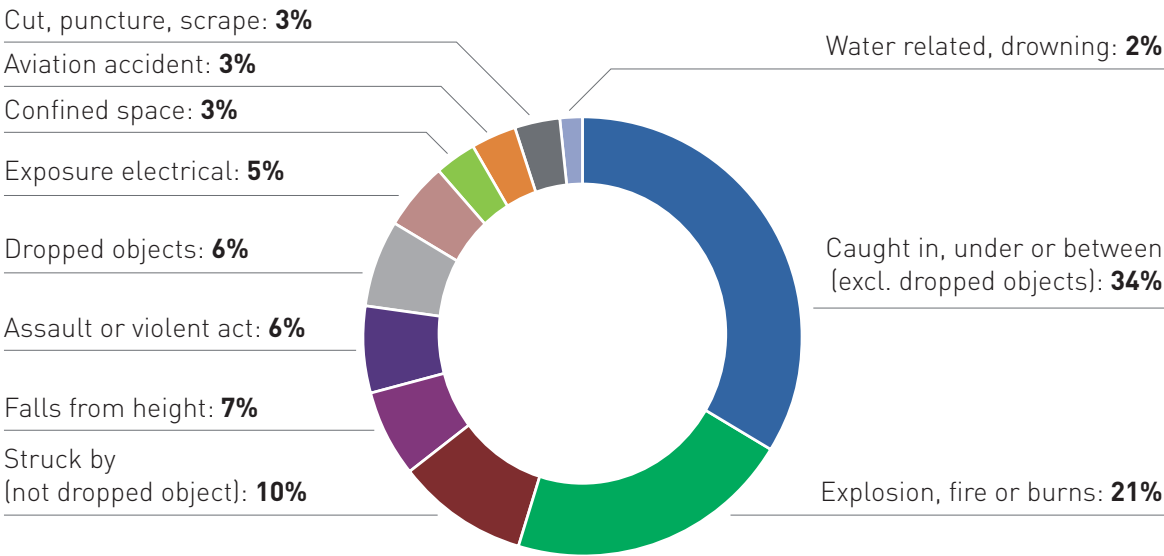


Figure 5: Fatalities and PI injuries by cause 2024

By activity, in 2024, 34% of the reported PI were categorized as 'drilling, workover, well operations' (23% in 2023, 38% in 2022), and 23% were categorized as 'maintenance, inspection, testing' (16% in 2023, 25% in 2022). 11% were categorized as 'construction, commissioning, decommissioning' (10% in 2023, 19% in 2022), and 8% were categorized as 'lifting, crane, rigging, deck operations' (13% in 2023, 6% in 2022).

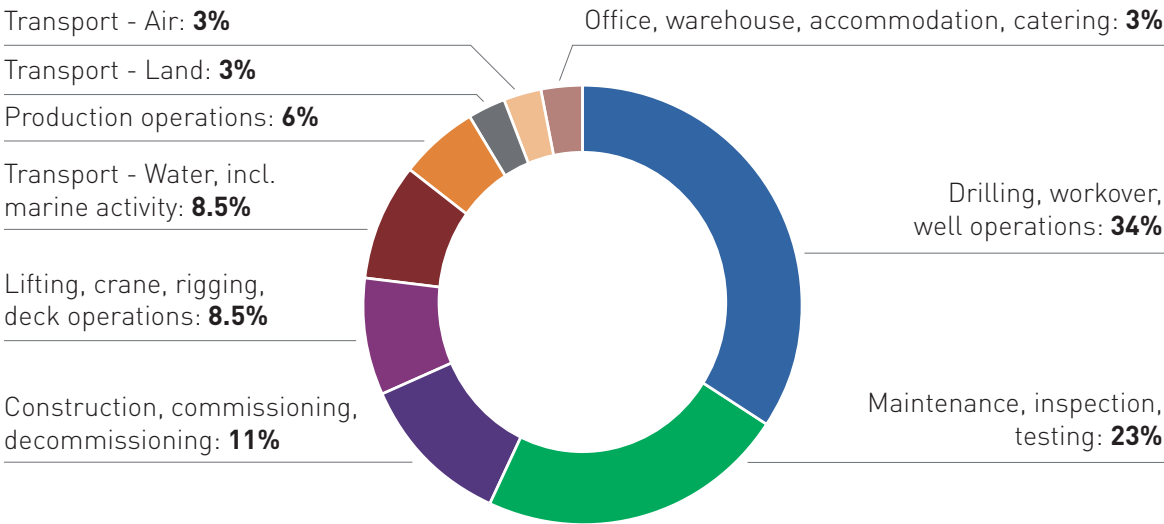


Figure 6: PI injuries by activity 2024

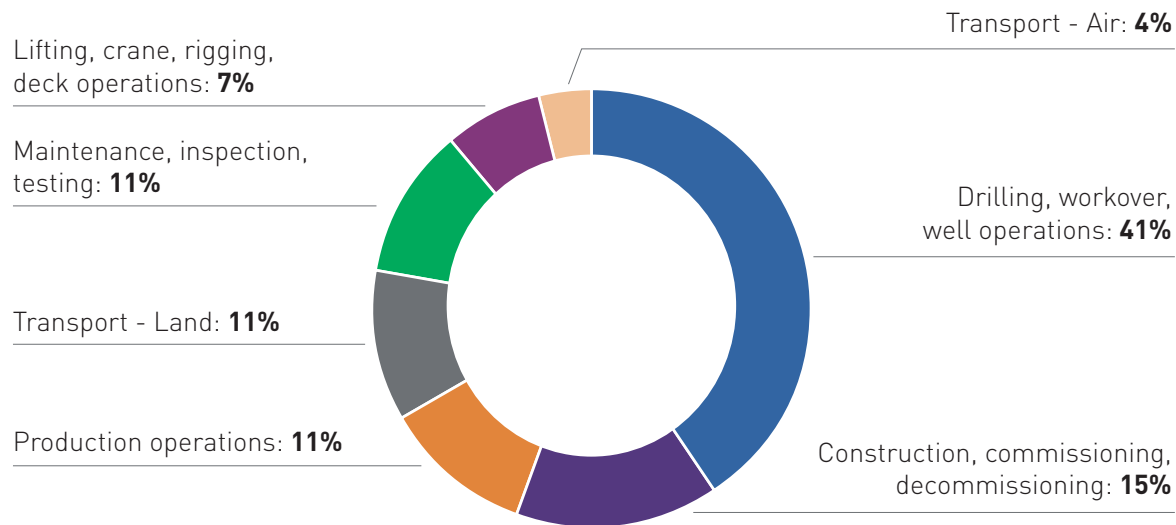


Figure 7: Fatalities by activity 2024

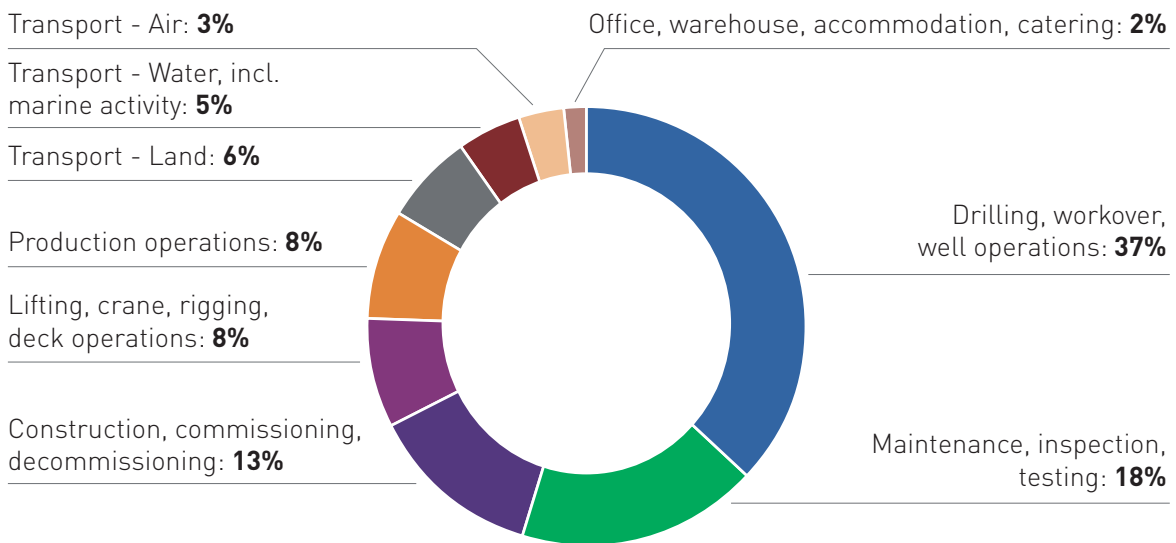


Figure 8: Fatalities and PI injuries by activity 2024

3.6 Fatalities and PI injuries by Life-Saving Rule

Table 9 shows the number of fatalities and PI injuries by Life-Saving Rule, aggregated for all years of reporting. Data are shown separately for fatality and PI injuries for each year of reporting in Appendix B, Table B6.

Table 9: Number of fatalities and permanent impairment injuries by Life-Saving Rule 2022-2024

Life-Saving Rule	Fatalities	Permanent impairment injuries	Total fatalities and permanent impairment injuries
Bypassing safety controls	4	3	7
Confined space	2	0	2
Driving	11	1	12
Energy isolation	7	11	18
Hot work	4	2	6
Line of fire	16	50	66
Safe mechanical lifting	5	3	8
Work authorization	5	0	5
Working at height	5	3	8
Other issue – no applicable rule	24	6	30
Unspecified	0	3	3
Total	83	82	165

Assessment of the applicability of the IOGP Life-Saving Rules to fatal and PI incident descriptions for 2022-2024 data has shown that at least 83% of the reported fatal incidents and 89% of the reported PI injuries are covered by the IOGP Life-Saving Rules and may have been prevented by the adoption of this system.

Analysis of fatality and PI data (2022-2024) reported to IOGP highlights that 19% of fatalities and 61% of PI injuries were assigned a primary Life-Saving Rule of line of fire, as shown in Figure 9.

Further analysis of fatality and PI data related to line of fire has been used to develop 10 main line of fire hazard categories and a package of resource materials focused on preventing line of fire-related fatalities and PI injuries. These are available at <https://www.iogp.org/life-savingrules/>.

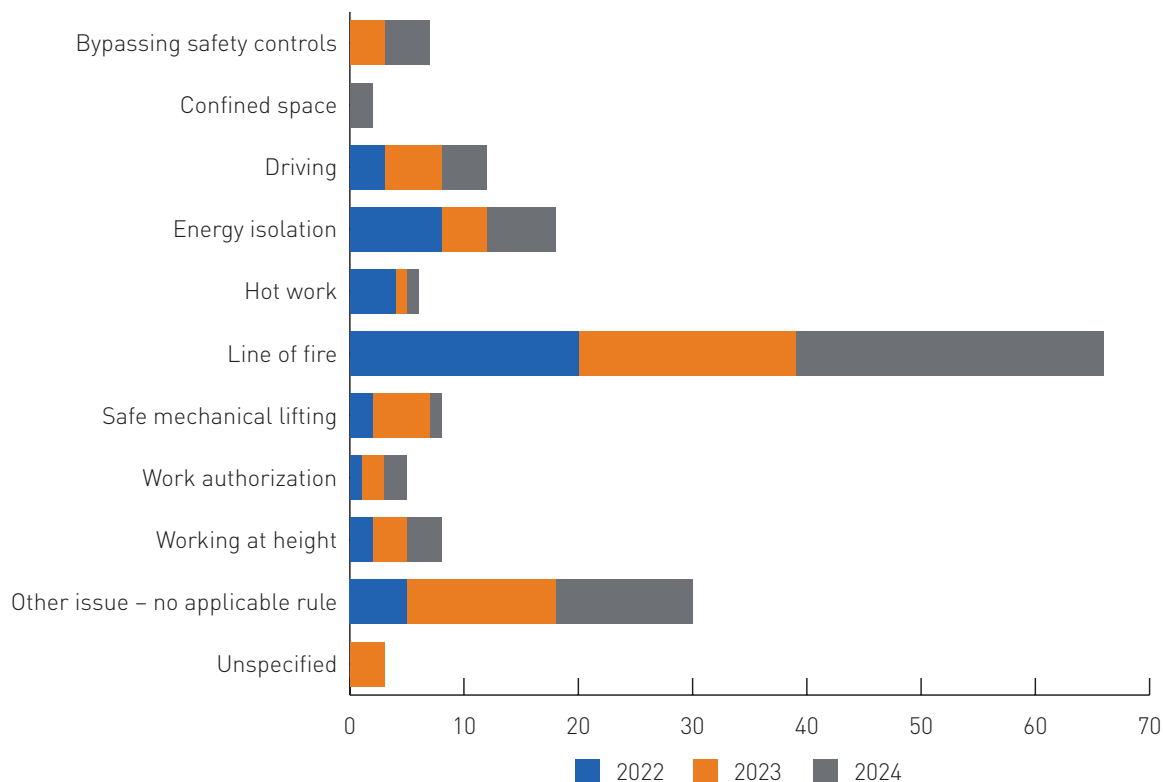


Figure 9: Fatalities and PI injuries by primary Life-Saving Rule 2022-2024

If the 'line of fire' IOGP Life-Saving Rule had been applied, 68% of the PI reported in 2024 might have been prevented (55% in 2023, 56% in 2022). Application of the 'energy isolation' Life-Saving Rule might have prevented 14% of the reported PI in 2024 (55% in 2023, 19% in 2022).

In 2024, 11 fatalities in 3 fatal incidents were assigned to 'Other issue – no applicable rule'.

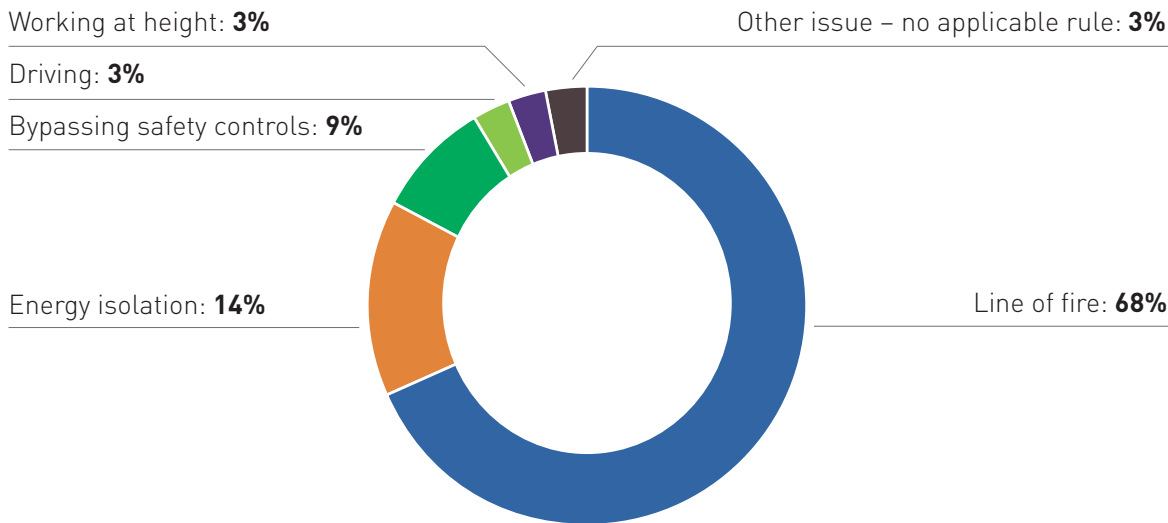


Figure 10: PI injuries by Life-Saving Rule 2024

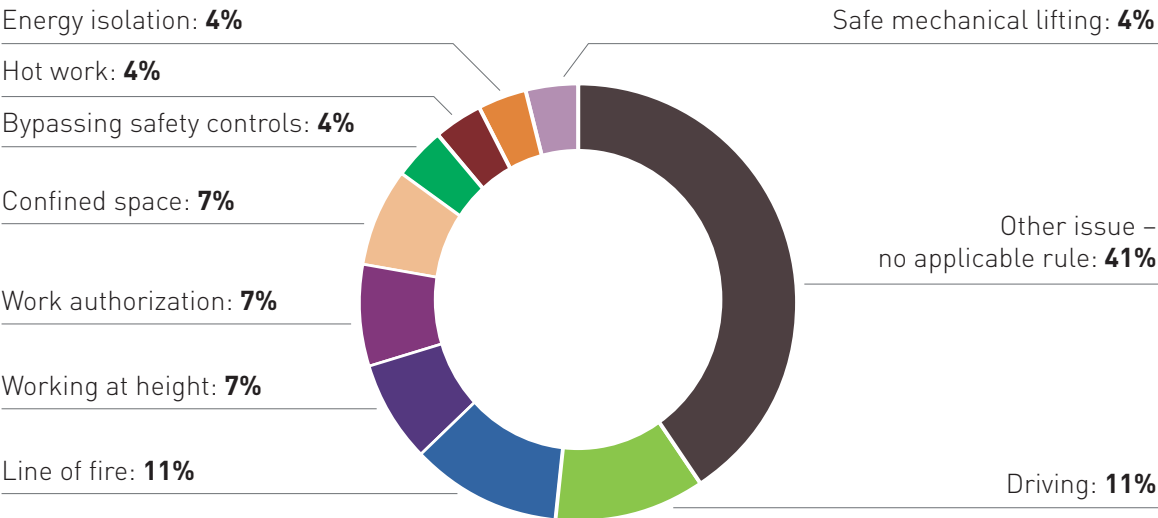


Figure 11: Fatalities by Life-Saving Rule 2024

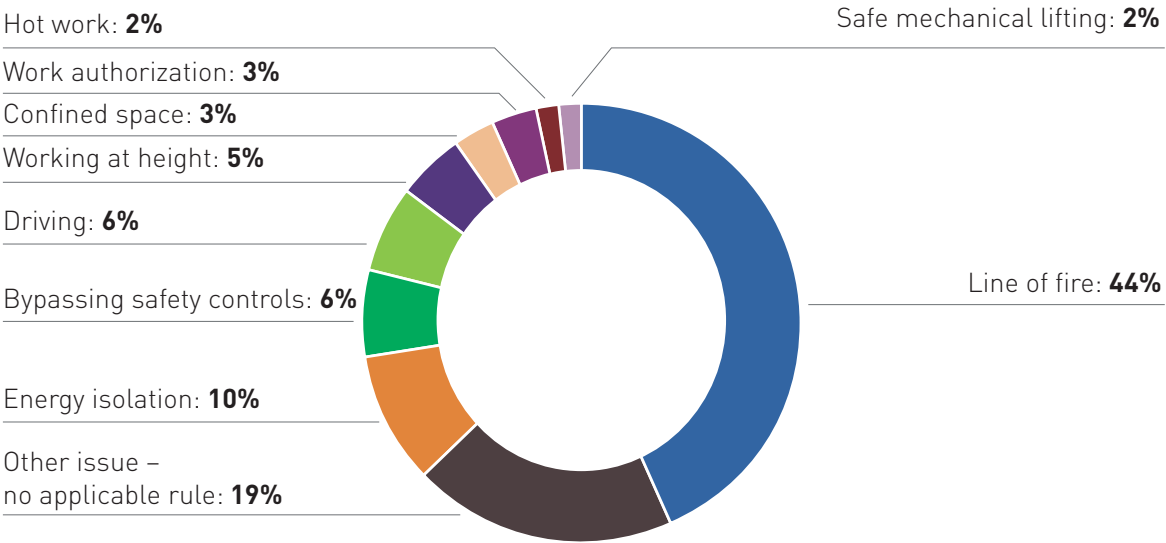


Figure 12: Fatalities and PI injuries by Life-Saving Rule 2024

For further Life-Saving Rules analysis related to fatalities refer to IOGP Report 2024s - *Safety Performance Indicators 2024*.

For more information on the IOGP Life-Saving Rules see IOGP Report 459 - *Life-Saving Rules* and <https://www.iogp.org/life-savingrules/>.

3.7 Fatalities and PI injuries by body part injured

Table 10 shows the number of fatalities and PI injuries by PI body part injured, aggregated for all years of reporting. Data are shown separately for fatality and PI injuries for each year of reporting in Appendix B, Table B7.

Table 10: Fatalities and permanent impairment injuries by body part 2022-2024

Body part	Fatalities	Permanent impairment injuries	Total fatalities and permanent impairment injuries
Arm/elbow/shoulder	0	7	7
Brain	0	1	1
Eye(s)	0	3	3
Feet/ankle	0	6	6
Finger(s)/thumb	0	36	36
Genitalia or other sexual organs	0	1	1
Hand/wrist	0	4	4
Head (incl. mouth, etc.)	25	6	31
Internal organs	22	4	26
Leg/knee/hip	3	6	9
Neck/torso/spine	10	5	15
Respiratory system	15	1	16
Toes	0	1	1
Unspecified	8	1	9

In 2024, 48% of reported PI affected the hand, wrist, or fingers (54% in 2023, 38% in 2022), 11% affected the head (incl. mouth, etc.) (7% in 2023, none in 2022), and 8% affected the neck/torso/spine (3% in 2023, 6% in 2022).

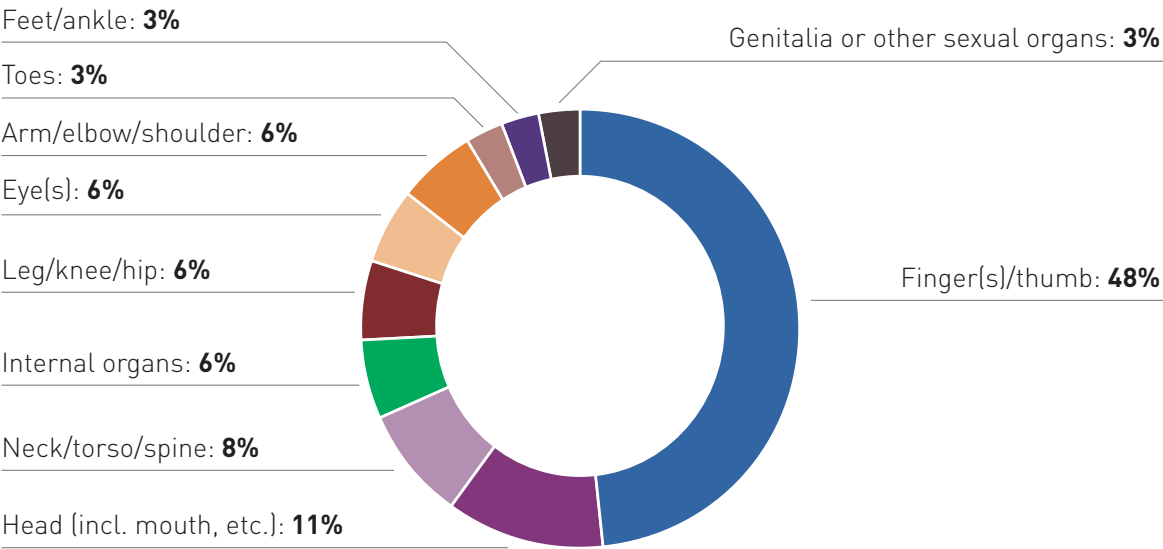


Figure 13: PI injuries by body part 2024

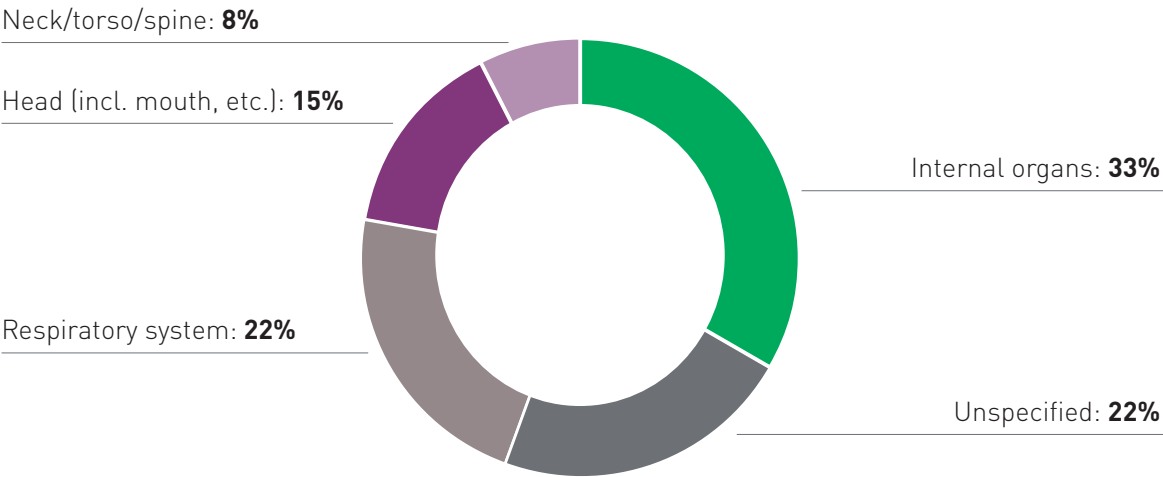


Figure 14: Fatalities by body part 2024

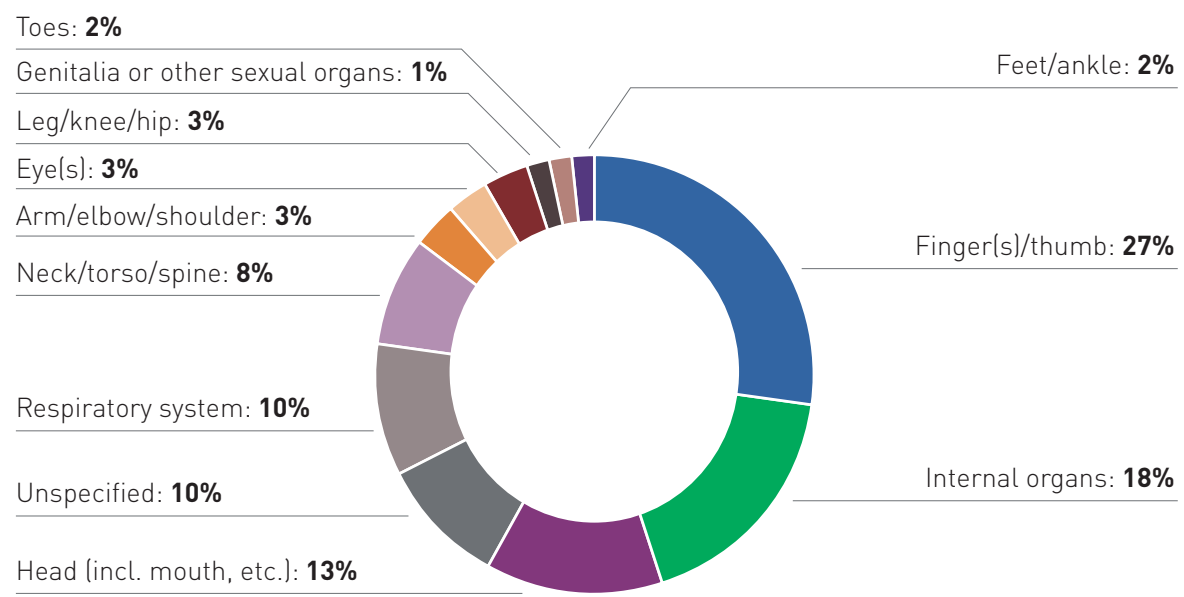


Figure 15: Fatalities and PI injuries by body part 2024

3.8 Fatalities and PI injuries by nature of injury

Table 11 shows the number of fatalities and PI injuries by nature of injury, aggregated for all years of reporting. Data are shown separately for each year of reporting in Appendix B, Table B8.

Table 11: Fatalities and permanent impairment injuries by nature of injury 2022-2024

Nature of injury	Fatalities	Permanent impairment injuries	Total fatalities and permanent impairment injuries
Amputation	1	35	36
Burn	15	5	20
Chemical or physical irritation or exposure	0	2	2
Concussion	0	1	1
Crushing	17	6	23
Cut, Puncture, Scrape	4	2	6
Electric shock	3	0	3
Fracture	1	18	19
Inhalation, poisoning, intoxication, asphyxiation, drowning	9	0	9
Loss of consciousness	1	0	1
Major/multiple system trauma	9	2	11
Organ function / loss or damage	2	3	5
Paralysis	0	1	1
Sensory loss/damage (e.g. hearing, sight, taste, smell)	0	3	3
Shot	1	0	1
Sprain, strain	0	2	2
Unspecified	20	2	22

In 2024, 51% of the PI were amputations with loss of bone inclusive of one or more joints (not able to be reattached) (39% in 2023, 31% in 2022), and 11% were burns (none in 2023, 6% in 2022).

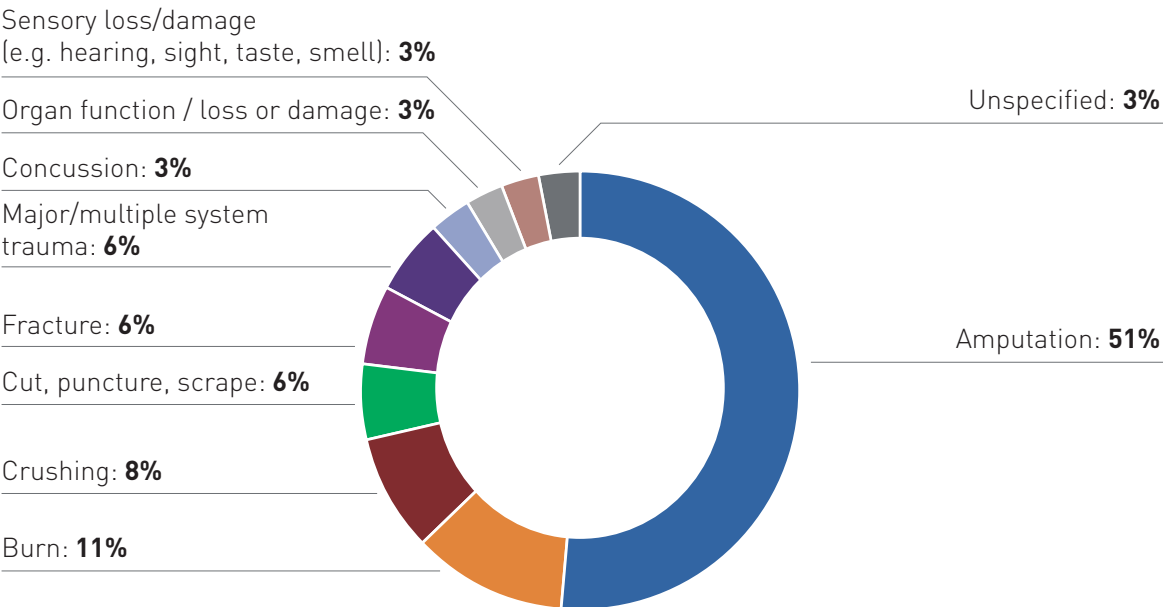


Figure 16: PI injuries by nature of injury 2024

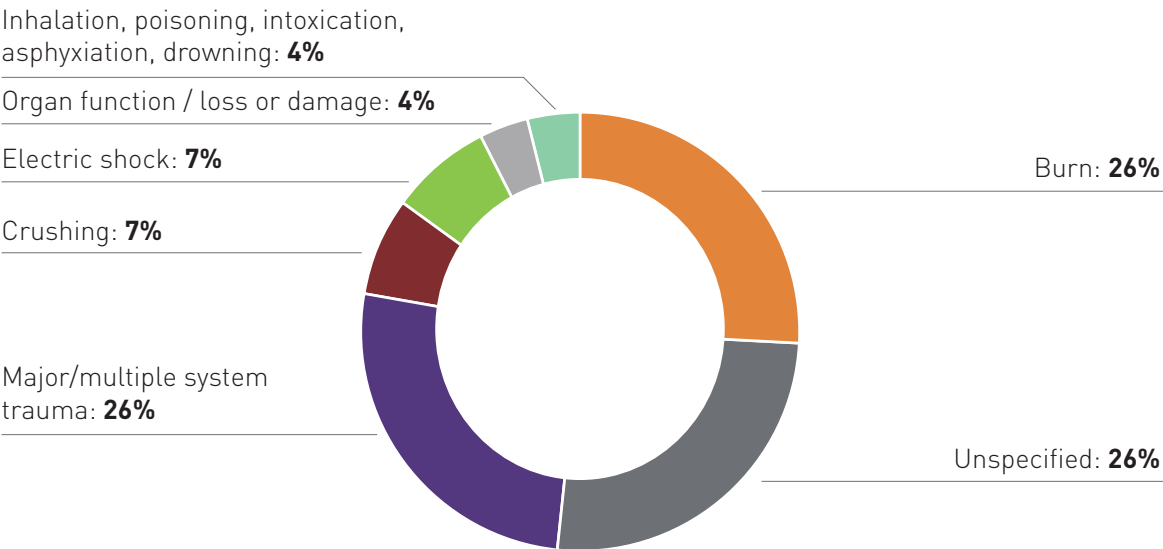


Figure 17: Fatalities by nature of injury 2024

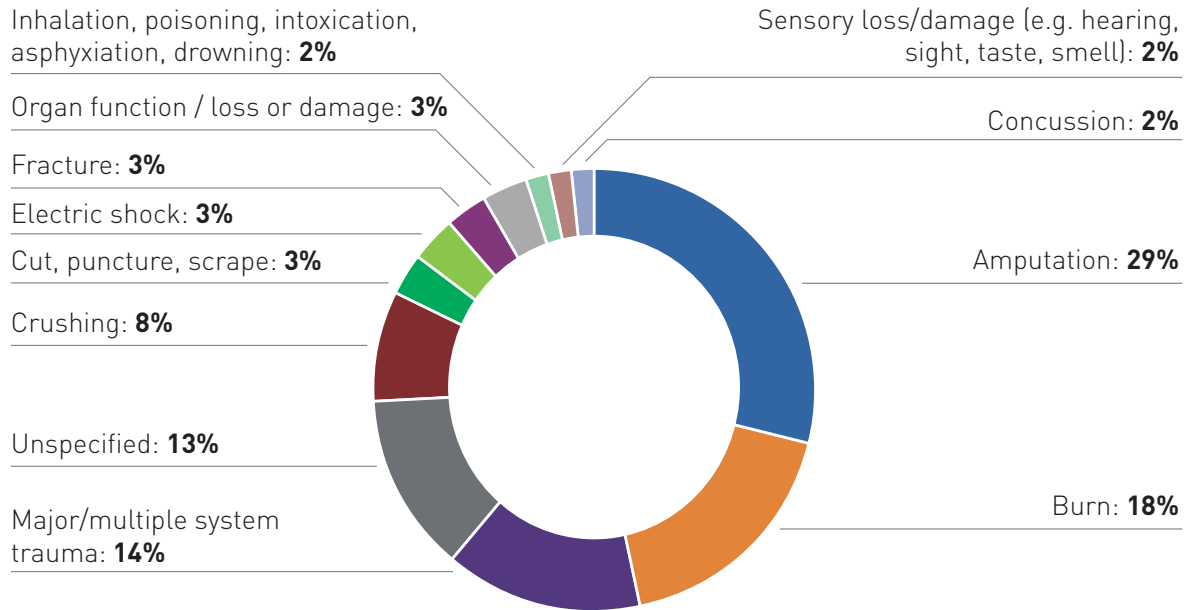


Figure 18: Fatalities and PI injuries by nature of injury 2024

3.9 PI injuries by PI category and subcategory

Table 12 shows the number of PI injuries by PI category and subcategory, aggregated for all years of reporting. Data are shown separately for fatality and PI injuries for each year of reporting in Appendix B, Table B9.

Table 12: Number of permanent impairment injuries by PI category and subcategory 2022-2024

Permanent impairment injuries	
Permanent loss of body parts	41
Amputations with loss of bone inclusive of one or more joints (not able to be reattached)	36
Loss of eye(s)	1
Loss of facial structures, example: nose, ears, leading to disfigurement	1
Loss of internal organs e.g. kidney, spleen (may also meet criteria 3.4.2)	2
Unspecified	1
Permanent reduction in psychological, social or cognitive function	4
Brain injury with ongoing impact to emotional regulation, speech, memory, or other cognitive dysfunction	2
Other	1
Unspecified	1

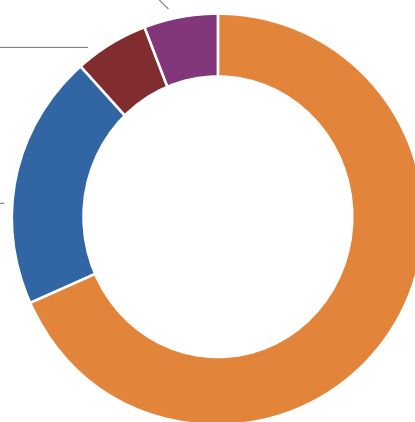
Permanent impairment injuries	
Permanent reduction in skin and/or musculoskeletal function	28
Burns: deep burns with scarring or skin grafts that impair body movement or other function	4
Fracture with ongoing impairment to mobility	14
Tendon/ligament tear with ongoing pain and/or inadequate joint range of movement for normal work or general function	2
Traumatic soft tissue injury to shoulder/knee/elbow with ongoing pain and/or inadequate joint range of movement	2
Other	3
Unspecified	3
Permanent reduction of organ function	7
Eye injury with loss of vision	2
Genital or other sexual organ injury leading to loss of function	1
Loss of hearing or persistent tinnitus (ear ringing) post injury after exposure to a blast	1
Loss of smell that creates a safety concern or significant personal impact	1
Lung injury, including inhalation, with impaired exertion tolerance/measurable loss of function	1
Spinal injury requiring use of wheelchair	1
Unspecified	2
Unspecified	2

In 2024, 68% were categorized as 'permanent loss of body parts' (36% in 2023, 37% in 2022), and 20% of reported PI were categorized as 'permanent reduction in skin and/or musculoskeletal function' (42% in 2023, 50% in 2022).

Permanent reduction in psychological, social or cognitive function: **6%**

Permanent reduction of organ function: **6%**

Permanent reduction in skin and/or musculoskeletal function: **20%**



Permanent loss of body parts: **68%**

Figure 19: PI injuries by PI category 2024

In 2024, 57% of reported PI were amputations with loss of bone inclusive of one or more joints (not able to be reattached) [36% in 2023, 31% in 2022], and 8% were deep burns with scarring or skin grafts that impair body movement or other function (none in 2023, 6% in 2022)

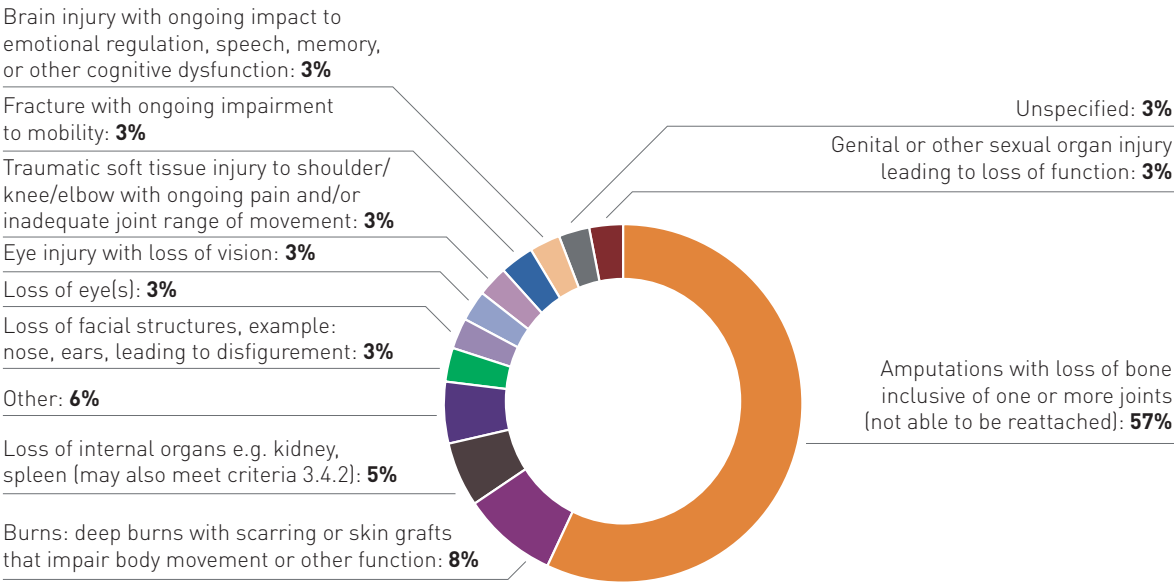


Figure 20: PI injuries by PI subcategory 2024

4. Results by region

In this Section, the safety performance of the contributing IOGP Member Companies is presented by region, as per Table 2 in Section 2 Scope of data submissions

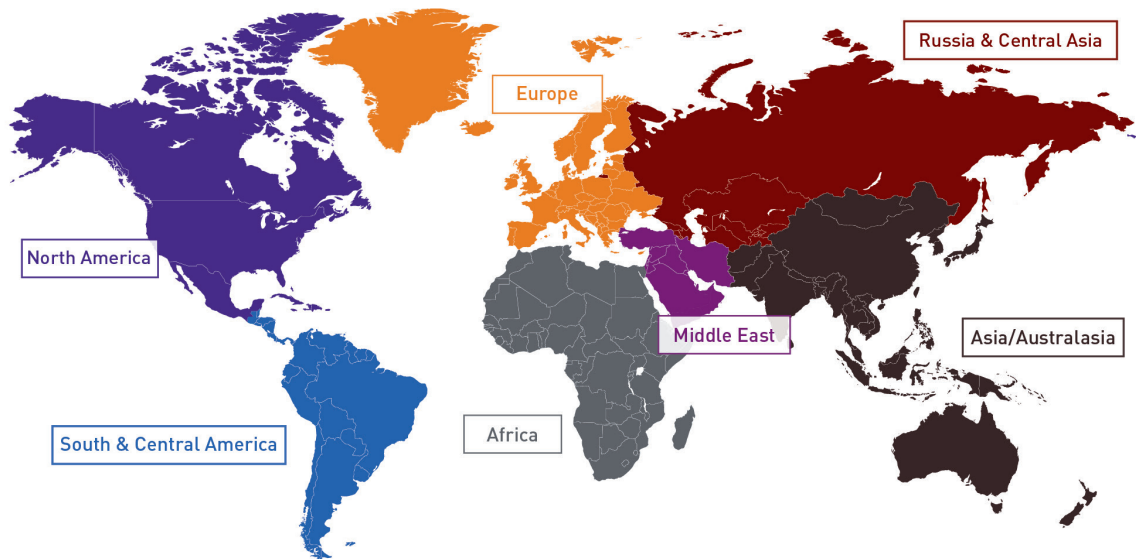


Figure 21: The 7 geographic regions

Countries and companies that are subject to international sanctions (Russia and Iran) are not featured in this Report.

Table 13 shows the number of fatalities and PI injuries by region, aggregated for all years of reporting. Work hours, fatalities and PI injuries are shown for each year of reporting in Appendix B, Table B10.

Table 13: Fatalities and PI injuries by region 2022-2024

Region*	Fatalities	Permanent impairment injuries	Total fatalities and permanent impairment injuries
Africa	15	6	21
Asia/Australasia	18	9	27
Europe	1	12	13
Middle East	24	17	41
North America	10	25	35
Russia & Central Asia	9	3	12
South & Central America	6	10	16

*Excludes Russia and Iran due to international sanctions

Rate calculations

Permanent impairment rate (PIR)

Number of company/contractor PI injuries per 100 million hours worked

Fatal accident rate (FAR)

The number of company/contractor fatalities* per 100 million hours worked

Fatality and permanent impairment rate (FPIR)

Number of company/contractor fatalities* plus PI injuries per 100 million hours

*In this Report, excludes fatalities where PI data (zero or greater) were not reported.

Figures 22 and 23 show PIR and FAR for all years of data, separately for all years of PI reporting.

Figure 24 shows FPIR for all years of data, separately for all years of PI reporting.

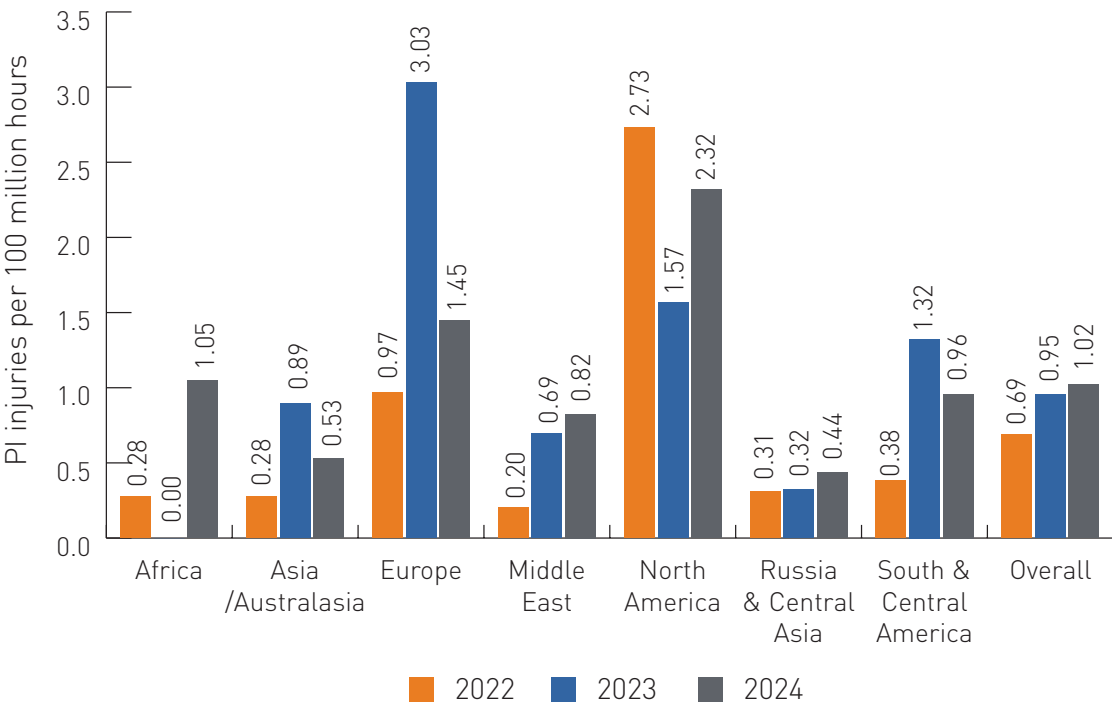


Figure 22: Permanent impairment rate by region 2022-2024

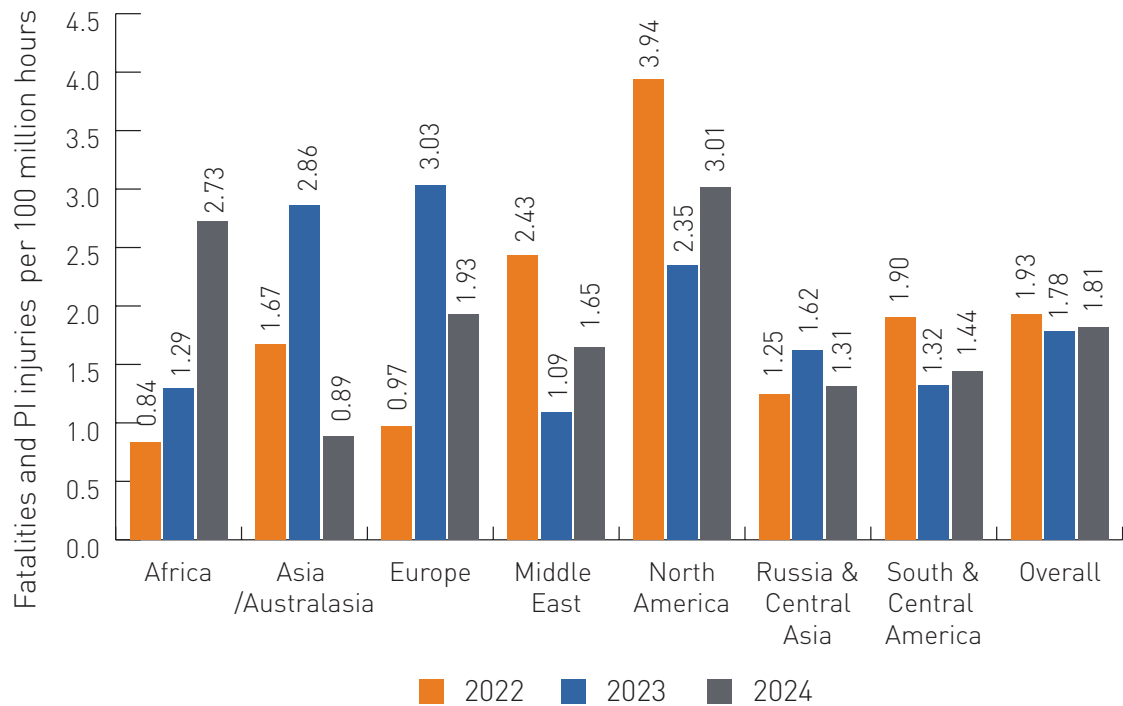


Figure 23: Fatality and permanent impairment rate by region 2022-2024

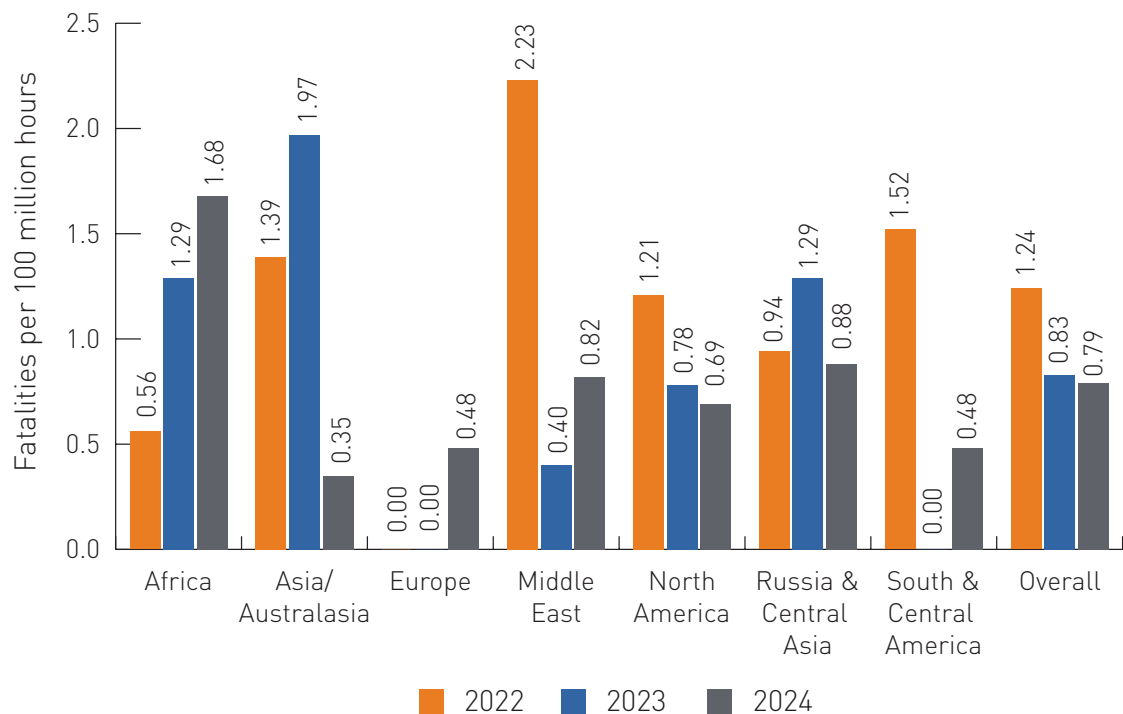


Figure 24: Fatal accident rate by region 2022-2024

5. Results by function

Table 14 shows the number of fatalities and PI injuries by work function, aggregated for all years of reporting, as per Table 2 in Section 2 Scope of data submissions. Data are shown separately for fatality and PI injuries for each year of reporting in Appendix B, Table B11.

Table 14: Fatalities and permanent impairment injuries by work function, 2022-2024

Work function	Fatalities	Permanent impairment injuries	Fatalities and permanent impairment injuries
Construction	11	12	23
Drilling	23	30	53
Exploration	0	1	1
Production	35	29	64
Unspecified	14	10	24

In 2024, 34% of reported PI were assigned to the drilling work function (36% in 2023, 44% in 2022), and 34% were also assigned to the production function (39% in 2023, 31% in 2022).

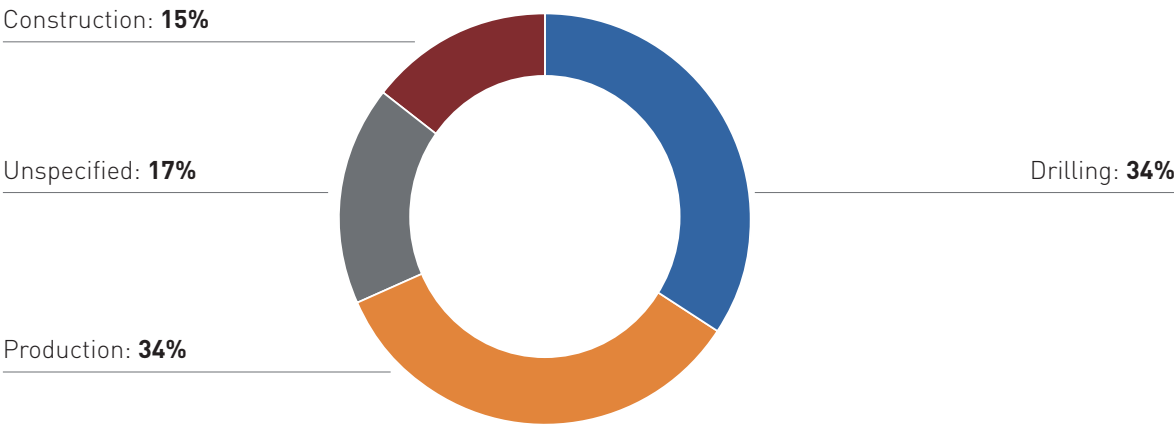


Figure 25: PI injuries by work function 2024

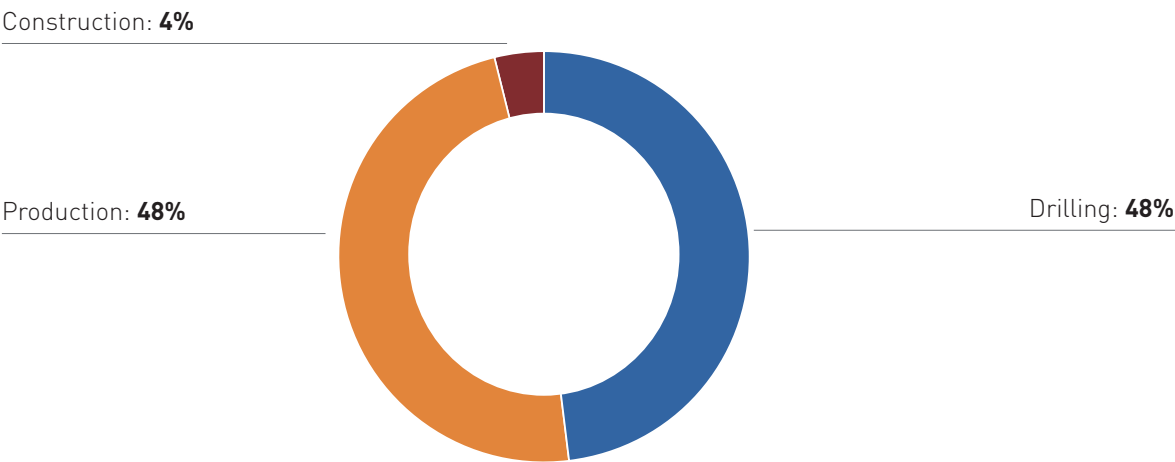


Figure 26: Fatalities by work function 2024

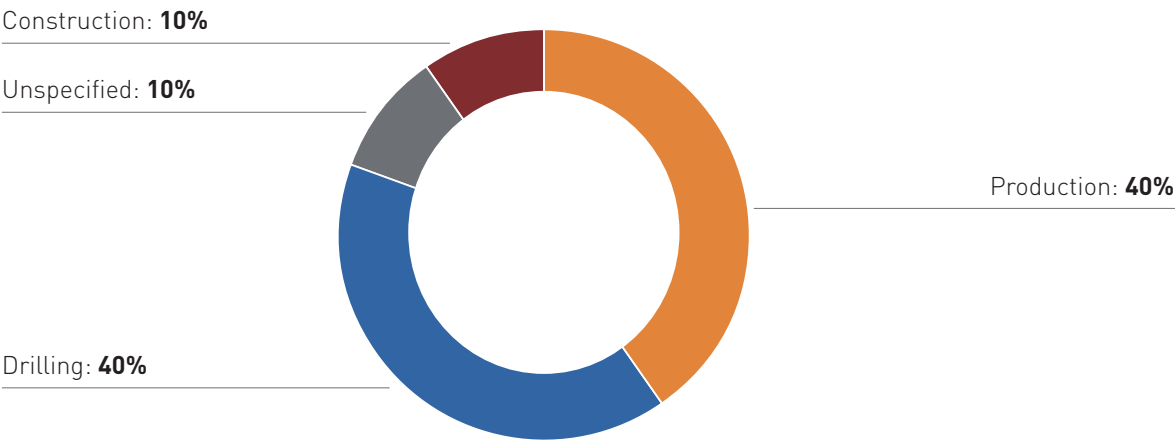


Figure 27: Fatalities and PI injuries by work function 2024

6. Results by company

Rate calculations

Permanent impairment rate (PIR)

Number of company/contractor PI injuries per 100 million hours worked

Fatal accident rate (FAR)

The number of company/contractor fatalities* per 100 million hours worked

Fatality and permanent impairment rate (FPIR)

Number of company/contractor fatalities* plus PI injuries per 100 million hours

*In this Report, excludes fatalities where PI data (zero or greater) were not reported.

This Section compares the safety performance of individual companies with each other and with their performance in previous years, as per Table 2 in Section 2 Scope of data submissions.

For reasons of anonymity, each of the companies that has contributed relevant data and is to be included in this analysis has been allocated a unique code letter (A to VV). These codes change every year in line with FPIR performance. 3 companies were excluded from the analysis of 2024 data because they did not report contractor data (4 companies were excluded in 2023 and 2 companies were excluded in 2022). Results for 48 of the 51 participating companies are therefore shown in this Section.

Aligned with the safety performance indicators report, only larger company data rates have been included in this Section. All company data is available in Appendix B, Table B12.

In Figures 28, 29 and 30, the FPIR, FAR, and PIR are presented for those companies that, with their contractors, reported more than 50 million work hours in 2024. 19 companies met this criterion.

In Figure 28, companies are presented in rank order of company-with-contractor FPIR. The analysis shows that:

- In 2024, 7 of the 19 companies with their contractors had a lower FPIR than the average for companies with contractors reporting more than 50 million work hours (1.44).
- 12 of the 19 companies suffered one or more fatalities.
- 7 of the 19 companies reported no fatalities.

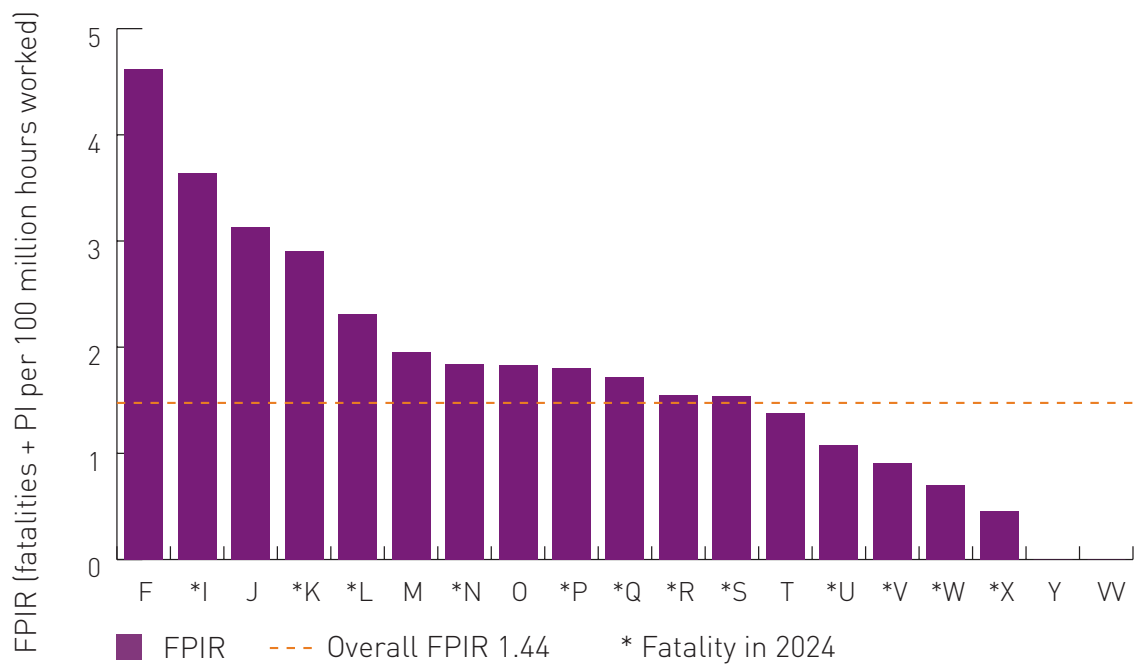


Figure 28: Fatality and permanent impairment rate (2024) – total workforce hours >50 million

In Figure 29, companies are presented in rank order of company-with-contractor FAR. The analysis shows that:

- In 2024, 11 of the 19 companies with their contractors had a lower FAR than the average for companies with contractors reporting more than 50 million work hours (0.47).

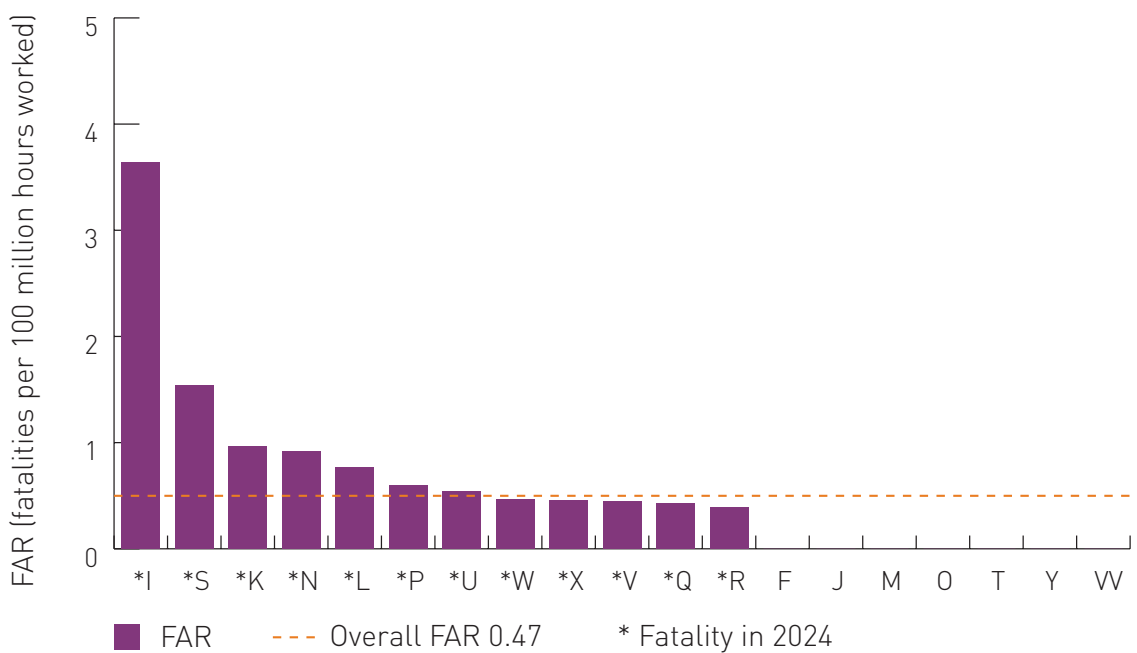


Figure 29: Fatal accident rate (2024) – total workforce hours >50 million

In Figure 30, companies are presented in rank order of company-with-contractor PIR. The analysis shows that:

- In 2024, 9 of the 19 companies with their contractors had a lower FPIR than the average for companies with contractors reporting more than 50 million work hours (0.97).

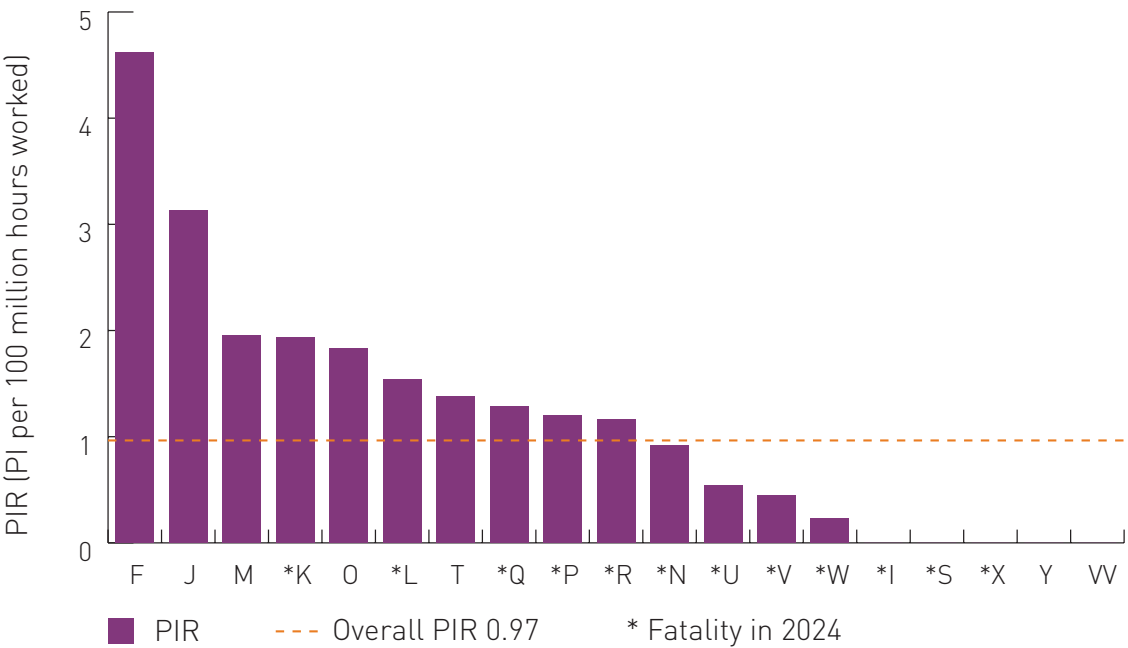


Figure 30: Permanent impairment rate (2024) – total workforce hours >50 million

Details for all companies are summarised in Table B12 and should be taken into consideration before conducting any comparative analysis.

Appendix A: Glossary of terms

Fatal accident rate (FAR)

The number of company/contractor fatalities per 100 million hours worked.

Fatality

A death as a result of a work-related incident. 'Delayed' deaths that occur within 180 days post incident are to be included if the death was a direct result of the incident (refer to the annual IOGP Safety Performance Indicators User Guide for the definition and further guidance on fatality, 'delayed' deaths, and work related injury).

Fatality and permanent impairment rate (FPIR)

Number of company/contractor fatalities plus permanent impairment injuries per 100 million hours.

OSHA

Occupational Safety and Health Administration.

Permanent Impairment (PI)

Permanent impairment is defined as the outcome of a work-related injury from which the worker cannot or is not expected to return to their previous (pre-incident) whole person function as a result of an acute, single incident, resulting in any of the following:

- Permanent loss of body parts
- Permanent reduction of organ's physiological function
- Permanent reduction in skin and musculoskeletal function
- Permanent reduction in psychological, social, or cognitive function

For reporting purposes, the classification is to be made within 180 days of the injury.

"Whole person function" relates to a person's physiological/psychological status. Impairment is defined as either, or both:

- Loss of ability to return to the previous work role
- Loss of ability to perform the activities of daily living (ADL)

Permanent impairment rate (PIR)

Number of company/contractor permanent impairment injuries per 100 million hours worked.

See IOGP Report 2024s - *IOGP Safety performance indicators – 2024 data* for full glossary.

Appendix B: Data tables

Note 2022 and 2023 data includes corrections to previously published data provided by participating companies.

Sections 2 and 3

Table B1: Summary of data 2022

Region	Data type	Operations	Hours worked (thousands)	Fatalities	PI injuries	PIR	FPIR	FAR
Africa	Company	Onshore	61,872	0	0	0.00	0.00	0.00
Africa	Company	Offshore	10,240	0	0	0.00	0.00	0.00
Africa	Contractor	Onshore	227,406	1	1	0.44	0.88	0.44
Africa	Contractor	Offshore	58,747	1	0	0.00	1.70	1.70
Asia/Australasia	Company	Onshore	58,251	0	0	0.00	0.00	0.00
Asia/Australasia	Company	Offshore	22,105	0	0	0.00	0.00	0.00
Asia/Australasia	Contractor	Onshore	169,626	5	1	0.59	3.54	2.95
Asia/Australasia	Contractor	Offshore	109,695	0	0	0.00	0.00	0.00
Europe	Company	Onshore	65,382	0	0	0.00	0.00	0.00
Europe	Company	Offshore	19,832	0	1	5.04	5.04	0.00
Europe	Contractor	Onshore	47,433	0	0	0.00	0.00	0.00
Europe	Contractor	Offshore	73,903	0	1	1.35	1.35	0.00
Middle East	Company	Onshore	54,749	2	0	0.00	3.65	3.65
Middle East	Company	Offshore	16,406	0	0	0.00	0.00	0.00
Middle East	Contractor	Onshore	300,998	5	1	0.33	1.99	1.66
Middle East	Contractor	Offshore	121,595	4	0	0.00	3.29	3.29
North America	Company	Onshore	68,185	0	0	0.00	0.00	0.00
North America	Company	Offshore	6,330	0	0	0.00	0.00	0.00
North America	Contractor	Onshore	222,723	4	9	4.04	5.84	1.80
North America	Contractor	Offshore	32,508	0	0	0.00	0.00	0.00
Russia & Central Asia	Company	Onshore	118,114	0	0	0.00	0.00	0.00
Russia & Central Asia	Company	Offshore	32,271	0	0	0.00	0.00	0.00
Russia & Central Asia	Contractor	Onshore	160,451	3	1	0.62	2.49	1.87
Russia & Central Asia	Contractor	Offshore	9,834	0	0	0.00	0.00	0.00

Region	Data type	Operations	Hours worked (thousands)	Fatalities	PI injuries	PIR	FPIR	FAR
South & Central America	Company	Onshore	39,422	0	0	0.00	0.00	0.00
South & Central America	Company	Offshore	15,408	0	0	0.00	0.00	0.00
South & Central America	Contractor	Onshore	107,294	2	1	0.93	2.80	1.86
South & Central America	Contractor	Offshore	100,373	2	0	0.00	1.99	1.99
Africa	SUBTOTAL	SUBTOTAL	358,265	2	1	0.28	0.84	0.56
Asia/Australasia	SUBTOTAL	SUBTOTAL	359,677	5	1	0.28	1.67	1.39
Europe	SUBTOTAL	SUBTOTAL	206,550	0	2	0.97	0.97	0.00
Middle East	SUBTOTAL	SUBTOTAL	493,748	11	1	0.20	2.43	2.23
North America	SUBTOTAL	SUBTOTAL	329,746	4	9	2.73	3.94	1.21
Russia & Central Asia	SUBTOTAL	SUBTOTAL	320,670	3	1	0.31	1.25	0.94
South & Central America	SUBTOTAL	SUBTOTAL	262,497	4	1	0.38	1.90	1.52
TOTAL	Company	Onshore	465,975	2	0	0.00	0.43	0.43
TOTAL	Company	Offshore	122,592	0	1	0.82	0.82	0.00
TOTAL	Contractor	Onshore	1,235,931	20	14	1.13	2.75	1.62
TOTAL	Contractor	Offshore	506,655	7	1	0.20	1.58	1.38
GRAND TOTAL	TOTAL	TOTAL	2,331,153	29	16	0.69	1.93	1.24

Table B2: Summary of data 2023

Region	Data type	Operations	Hours worked (thousands)	Fatalities	PI injuries	PIR	FPIR	FAR
Africa	Company	Onshore	65,070	0	0	0.00	0.00	0.00
Africa	Company	Offshore	10,652	0	0	0.00	0.00	0.00
Africa	Contractor	Onshore	239,103	4	0	0.00	1.67	1.67
Africa	Contractor	Offshore	71,560	1	0	0.00	1.40	1.40
Asia/Australasia	Company	Onshore	135,664	0	2	1.47	1.47	0.00
Asia/Australasia	Company	Offshore	48,406	0	0	0.00	0.00	0.00
Asia/Australasia	Contractor	Onshore	190,341	9	2	1.05	5.78	4.73
Asia/Australasia	Contractor	Offshore	184,978	2	1	0.54	1.62	1.08
Europe	Company	Onshore	74,836	0	1	1.34	1.34	0.00
Europe	Company	Offshore	21,885	0	0	0.00	0.00	0.00
Europe	Contractor	Onshore	62,669	0	0	0.00	0.00	0.00
Europe	Contractor	Offshore	71,620	0	6	8.38	8.38	0.00
Middle East	Company	Onshore	143,761	0	1	0.70	0.70	0.00
Middle East	Company	Offshore	27,347	0	0	0.00	0.00	0.00
Middle East	Contractor	Onshore	678,487	4	5	0.74	1.33	0.59

Region	Data type	Operations	Hours worked (thousands)	Fatalities	PI injuries	PIR	FPIR	FAR
Middle East	Contractor	Offshore	157,621	0	1	0.63	0.63	0.00
North America	Company	Onshore	79,902	2	0	0.00	2.50	2.50
North America	Company	Offshore	8,535	0	0	0.00	0.00	0.00
North America	Contractor	Onshore	258,911	1	5	1.93	2.32	0.39
North America	Contractor	Offshore	35,905	0	1	2.79	2.79	0.00
Russia & Central Asia	Company	Onshore	139,692	2	1	0.72	2.15	1.43
Russia & Central Asia	Company	Offshore	43,811	2	0	0.00	4.57	4.57
Russia & Central Asia	Contractor	Onshore	114,351	0	0	0.00	0.00	0.00
Russia & Central Asia	Contractor	Offshore	11,267	0	0	0.00	0.00	0.00
South & Central America	Company	Onshore	51,861	0	0	0.00	0.00	0.00
South & Central America	Company	Offshore	17,415	0	0	0.00	0.00	0.00
South & Central America	Contractor	Onshore	169,086	0	1	0.59	0.59	0.00
South & Central America	Contractor	Offshore	139,310	0	4	2.87	2.87	0.00
Africa	SUBTOTAL	SUBTOTAL	386,385	5	0	0.00	1.29	1.29
Asia/Australasia	SUBTOTAL	SUBTOTAL	559,390	11	5	0.89	2.86	1.97
Europe	SUBTOTAL	SUBTOTAL	231,011	0	7	3.03	3.03	0.00
Middle East	SUBTOTAL	SUBTOTAL	1,007,216	4	7	0.69	1.09	0.40
North America	SUBTOTAL	SUBTOTAL	383,252	3	6	1.57	2.35	0.78
Russia & Central Asia	SUBTOTAL	SUBTOTAL	309,121	4	1	0.32	1.62	1.29
South & Central America	SUBTOTAL	SUBTOTAL	377,672	0	5	1.32	1.32	0.00
TOTAL	Company	Onshore	690,786	4	5	0.72	1.30	0.58
TOTAL	Company	Offshore	178,051	2	0	0.00	1.12	1.12
TOTAL	Contractor	Onshore	1,712,948	18	13	0.76	1.81	1.05
TOTAL	Contractor	Offshore	672,261	3	13	1.93	2.38	0.45
GRAND TOTAL	TOTAL	TOTAL	3,254,047	27	31	0.95	1.78	0.83

Table B3: Summary of data 2024

Region	Data type	Operations	Hours worked (thousands)	Fatalities	PI injuries	PIR	FPIR	FAR
Africa	Company	Onshore	71,659	0	0	0.00	0.00	0.00
Africa	Company	Offshore	15,387	0	0	0.00	0.00	0.00
Africa	Contractor	Onshore	319,815	2	4	1.25	1.88	0.63
Africa	Contractor	Offshore	69,721	6	1	1.43	10.04	8.61
Asia/Australasia	Company	Onshore	133,791	0	0	0.00	0.00	0.00
Asia/Australasia	Company	Offshore	43,362	0	0	0.00	0.00	0.00

Region	Data type	Operations	Hours worked (thousands)	Fatalities	PI injuries	PIR	FPIR	FAR
Asia/Australasia	Contractor	Onshore	207,446	0	2	0.96	0.96	0.00
Asia/Australasia	Contractor	Offshore	179,607	2	1	0.56	1.67	1.11
Europe	Company	Onshore	61,459	0	1	1.63	1.63	0.00
Europe	Company	Offshore	21,579	1	1	4.63	9.27	4.63
Europe	Contractor	Onshore	47,913	0	0	0.00	0.00	0.00
Europe	Contractor	Offshore	76,621	0	1	1.31	1.31	0.00
Middle East	Company	Onshore	121,679	2	1	0.82	2.47	1.64
Middle East	Company	Offshore	21,354	0	1	4.68	4.68	0.00
Middle East	Contractor	Onshore	785,297	6	7	0.89	1.66	0.76
Middle East	Contractor	Offshore	162,606	1	0	0.00	0.61	0.61
North America	Company	Onshore	85,125	0	0	0.00	0.00	0.00
North America	Company	Offshore	7,183	0	0	0.00	0.00	0.00
North America	Contractor	Onshore	306,270	3	9	2.94	3.92	0.98
North America	Contractor	Offshore	33,153	0	1	3.02	3.02	0.00
Russia & Central Asia	Company	Onshore	86,937	1	1	1.15	2.30	1.15
Russia & Central Asia	Company	Offshore	18,262	1	0	0.00	5.48	5.48
Russia & Central Asia	Contractor	Onshore	112,574	0	0	0.00	0.00	0.00
Russia & Central Asia	Contractor	Offshore	10,416	0	0	0.00	0.00	0.00
South & Central America	Company	Onshore	57,690	0	0	0.00	0.00	0.00
South & Central America	Company	Offshore	18,721	0	0	0.00	0.00	0.00
South & Central America	Contractor	Onshore	193,958	1	2	1.03	1.55	0.52
South & Central America	Contractor	Offshore	147,052	1	2	1.36	2.04	0.68
Africa	SUBTOTAL	SUBTOTAL	476,583	8	5	1.05	2.73	1.68
Asia/Australasia	SUBTOTAL	SUBTOTAL	564,206	2	3	0.53	0.89	0.35
Europe	SUBTOTAL	SUBTOTAL	207,573	1	3	1.45	1.93	0.48
Middle East	SUBTOTAL	SUBTOTAL	1,090,936	9	9	0.82	1.65	0.82
North America	SUBTOTAL	SUBTOTAL	431,730	3	10	2.32	3.01	0.69
Russia & Central Asia	SUBTOTAL	SUBTOTAL	228,189	2	1	0.44	1.31	0.88
South & Central America	SUBTOTAL	SUBTOTAL	417,421	2	4	0.96	1.44	0.48
TOTAL	Company	Onshore	618,340	3	3	0.49	0.97	0.49
TOTAL	Company	Offshore	145,849	2	2	1.37	2.74	1.37
TOTAL	Contractor	Onshore	1,973,274	12	24	1.22	1.82	0.61
TOTAL	Contractor	Offshore	679,176	10	6	0.88	2.36	1.47
GRAND TOTAL	TOTAL	TOTAL	3,416,638	27	35	1.02	1.81	0.79

Table B4: Fatalities and permanent impairment injuries by cause

Cause	Fatalities					PI injuries					FPI incidents				
	Company	Contractor	OVERALL	Onshore	Offshore	Company	Contractor	OVERALL	Onshore	Offshore	Company	Contractor	OVERALL	Onshore	Offshore
2022															
Assault or violent act	0	1	1	1	0	0	0	0	0	0	0	1	1	1	0
Aviation accident	0	1	1	0	1	0	0	0	0	0	0	1	1	0	1
Caught in, under or between (excl. dropped objects)	0	5	5	3	2	0	5	5	5	0	0	9	9	8	1
Confined space	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Cut, puncture, scrape	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Dropped objects	0	6	6	4	2	0	3	3	2	1	0	9	9	6	3
Explosion, fire or burns	0	3	3	3	0	0	0	0	0	0	0	2	2	2	0
Exposure electrical	0	0	0	0	0	0	1	1	1	0	0	1	1	1	0
Exposure noise, chemical, biological, vibration, extreme temperature	0	1	1	0	1	0	1	1	1	0	0	2	2	1	1
Falls from height	2	1	3	2	1	0	0	0	0	0	1	1	2	1	1
Overexertion, strain	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Pressure release	0	3	3	3	0	1	3	4	3	1	1	4	5	4	1
Slips and trips (at same height)	0	0	0	0	0	0	1	1	1	0	0	1	1	1	0
Struck by (not dropped object)	0	6	6	6	0	0	1	1	1	0	0	7	7	7	0
Water related, drowning	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Unspecified - Other	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Safety performance indicators – Fatality and permanent impairment injuries (FPI) – 2024 data

Cause	Fatalities					PI injuries					FPI incidents				
	Company	Contractor	OVERALL	Onshore	Offshore	Company	Contractor	OVERALL	Onshore	Offshore	Company	Contractor	OVERALL	Onshore	Offshore
2023															
Assault or violent act	0	8	8	8	0	0	0	0	0	0	0	2	2	2	0
Aviation accident	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Caught in, under or between (excl. dropped objects)	2	2	4	3	1	1	12	13	6	7	2	14	16	8	8
Confined space	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Cut, puncture, scrape	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Dropped objects	1	0	1	1	0	0	2	2	1	1	1	2	3	2	1
Explosion, fire or burns	0	5	5	4	1	0	1	1	1	0	0	4	4	3	1
Exposure electrical	0	1	1	1	0	0	0	0	0	0	0	1	1	1	0
Exposure noise, chemical, biological, vibration, extreme temperature	1	1	2	2	0	0	0	0	0	0	1	1	2	2	0
Falls from height	0	1	1	1	0	0	1	1	1	0	0	2	2	2	0
Overexertion, strain	0	0	0	0	0	1	1	2	1	1	1	1	2	1	1
Pressure release	0	1	1	1	0	0	1	1	0	1	0	2	2	1	1
Slips and trips (at same height)	0	0	0	0	0	0	1	1	0	1	0	1	1	0	1
Struck by (not dropped object)	0	2	2	1	1	1	7	8	6	2	1	9	10	7	3
Water related, drowning	2	0	2	0	2	1	0	1	1	0	2	0	2	1	1
Unspecified - Other	0	0	0	0	0	1	0	1	1	0	1	0	1	1	0

Safety performance indicators – Fatality and permanent impairment injuries (FPI) – 2024 data

Cause	Fatalities					PI injuries					FPI incidents				
	Company	Contractor	OVERALL	Onshore	Offshore	Company	Contractor	OVERALL	Onshore	Offshore	Company	Contractor	OVERALL	Onshore	Offshore
2024															
Assault or violent act	0	4	4	4	0	0	0	0	0	0	0	1	1	1	0
Aviation accident	1	0	1	0	1	1	0	1	0	1	1	0	1	0	1
Caught in, under or between (excl. dropped objects)	0	2	2	2	0	1	18	19	15	4	1	20	21	17	4
Confined space	0	1	1	0	1	0	1	1	0	1	0	2	2	0	2
Cut, puncture, scrape	0	0	0	0	0	0	2	2	2	0	0	2	2	2	0
Dropped objects	1	0	1	1	0	0	3	3	3	0	1	3	4	4	0
Explosion, fire or burns	1	8	9	0	9	1	3	4	3	1	2	4	6	2	4
Exposure electrical	0	2	2	1	1	0	1	1	1	0	0	3	3	2	1
Exposure noise, chemical, biological, vibration, extreme temperature	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Falls from height	0	3	3	3	0	1	0	1	1	0	1	3	4	4	0
Overexertion, strain	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Pressure release	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Slips and trips (at same height)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Struck by (not dropped object)	1	2	3	3	0	1	2	3	2	1	2	4	6	5	1
Water related, drowning	1	0	1	1	0	0	0	0	0	0	1	0	1	1	0
Unspecified - Other	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Table B5: Fatalities and permanent impairment injuries by activity

Activity	Fatalities					PI injuries					FPI incidents				
	Company	Contractor	OVERALL	Onshore	Offshore	Company	Contractor	OVERALL	Onshore	Offshore	Company	Contractor	OVERALL	Onshore	Offshore
2022															
Construction, commissioning, decommissioning	0	4	4	3	1	1	2	3	2	1	1	4	5	3	2
Diving (incl. decompression), subsea, ROV	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Drilling, workover, well operations	2	7	9	8	1	0	6	6	6	0	1	13	14	13	1
Excavation, trenching, ground disturbance	0	1	1	1	0	0	0	0	0	0	0	1	1	1	0
Lifting, crane, rigging, deck operations	0	4	4	3	1	0	1	1	1	0	0	5	5	4	1
Maintenance, inspection, testing	0	3	3	2	1	0	4	4	3	1	0	7	7	5	2
Office, warehouse, accommodation, catering	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Production operations	0	3	3	3	0	0	1	1	1	0	0	3	3	3	0
Seismic/survey operations	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Transport - Air	0	1	1	0	1	0	0	0	0	0	0	1	1	0	1
Transport - Land	0	3	3	1	2	0	0	0	0	0	0	2	2	1	1
Transport - Water, incl. marine activity	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Unspecified - other	0	1	1	1	0	0	1	1	1	0	0	2	2	2	0

Safety performance indicators – Fatality and permanent impairment injuries (FPI) – 2024 data

Activity	Fatalities					PI injuries					FPI incidents				
	Company	Contractor	OVERALL	Onshore	Offshore	Company	Contractor	OVERALL	Onshore	Offshore	Company	Contractor	OVERALL	Onshore	Offshore
2023															
Construction, commissioning, decommissioning	0	3	3	3	0	0	3	3	1	2	0	6	6	4	2
Diving (incl. decompression), subsea, ROV	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Drilling, workover, well operations	0	3	3	2	1	1	6	7	6	1	1	9	10	8	2
Excavation, trenching, ground disturbance	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Lifting, crane, rigging, deck operations	2	0	2	0	2	0	4	4	2	2	1	4	5	2	3
Maintenance, inspection, testing	1	3	4	3	1	1	4	5	3	2	2	6	8	5	3
Office, warehouse, accommodation, catering	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Production operations	0	7	7	7	0	1	4	5	3	2	1	6	7	5	2
Seismic/survey operations	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Transport - Air	0	0	0	0	0	1	0	1	1	0	1	0	1	1	0
Transport - Land	3	4	7	7	0	0	0	0	0	0	2	2	4	4	0
Transport - Water, incl. marine activity	0	1	1	0	1	0	5	5	1	4	0	6	6	1	5
Unspecified - other	0	0	0	0	0	1	0	1	1	0	1	0	1	1	0

Safety performance indicators – Fatality and permanent impairment injuries (FPI) – 2024 data

Activity	Fatalities					PI injuries					FPI incidents				
	Company	Contractor	OVERALL	Onshore	Offshore	Company	Contractor	OVERALL	Onshore	Offshore	Company	Contractor	OVERALL	Onshore	Offshore
2024															
Construction, commissioning, decommissioning	0	4	4	4	0	0	4	4	4	0	0	5	5	5	0
Diving (incl. decompression), subsea, ROV	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Drilling, workover, well operations	2	9	11	4	7	1	11	12	11	1	3	14	17	14	3
Excavation, trenching, ground disturbance	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Lifting, crane, rigging, deck operations	1	1	2	2	0	0	3	3	3	0	1	4	5	5	0
Maintenance, inspection, testing	0	3	3	2	1	1	7	8	5	3	1	10	11	7	4
Office, warehouse, accommodation, catering	0	0	0	0	0	0	1	1	1	0	0	1	1	1	0
Production operations	0	3	3	0	3	1	1	2	1	1	1	3	4	1	3
Seismic/survey operations	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Transport - Air	1	0	1	0	1	1	0	1	0	1	1	0	1	0	1
Transport - Land	1	2	3	3	0	1	0	1	1	0	2	2	4	4	0
Transport - Water, incl. marine activity	0	0	0	0	0	0	3	3	1	2	0	3	3	1	2
Unspecified - other	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Table B6: Fatalities and permanent impairment injuries by Life-Saving Rule

Life-Saving Rule	Fatalities	Permanent impairment injuries
2022		
Driving	3	0
Energy isolation	5	3
Hot work	3	1
Line of fire	11	9
Safe mechanical lifting	2	0
Work authorization	1	0
Working at height	1	1
Other issue – no applicable rule	3	2
2023		
Bypassing safety controls	3	0
Driving	5	0
Energy isolation	1	3
Hot work	0	1
Line of fire	2	17
Safe mechanical lifting	2	3
Work authorization	2	0
Working at height	2	1
Other issue – no applicable rule	10	3
Unspecified	0	3
2024		
Bypassing safety controls	1	3
Confined space	2	0
Driving	3	1
Energy isolation	1	5
Hot work	1	0
Line of fire	3	24
Safe mechanical lifting	1	0
Work authorization	2	0
Working at height	2	1
Other issue – no applicable rule	11	1

Note: for categories not listed, values are zero. Refer to *IOGP Report 2024su - Safety Performance Indicators User Guide*, for a full list of options.

Table B7: Fatalities and permanent impairment injuries by body part

Body part	Fatalities	Permanent impairment injuries
2022		
Arm/elbow/shoulder	0	2
Feet/ankle	0	3
Finger(s)/thumb	0	4
Hand/wrist	0	2
Head (incl. mouth, etc.)	15	0
Internal organs	6	2
Leg/knee/hip	2	1
Neck/torso/spine	4	1
Respiratory system	2	1
2023		
Arm/elbow/shoulder	0	3
Brain	0	1
Eye(s)	0	1
Feet/ankle	0	2
Finger(s)/thumb	0	15
Hand/wrist	0	2
Head (incl. mouth, etc.)	6	2
Internal organs	7	0
Leg/knee/hip	1	3
Neck/torso/spine	4	1
Respiratory system	7	0
Unspecified	2	1
2024		
Arm/elbow/shoulder	0	2
Eye(s)	0	2
Feet/ankle	0	1
Finger(s)/thumb	0	17
Genitalia or other sexual organs	0	1
Head (incl. mouth, etc.)	4	4
Internal organs	9	2
Leg/knee/hip	0	2
Neck/torso/spine	2	3
Respiratory system	6	0
Toes	0	1
Unspecified	6	0

Note: for categories not listed, values are zero. Refer to *IOGP Report 2024su - Safety Performance Indicators User Guide*, for a full list of options.

Table B8: Fatalities and permanent impairment injuries by nature of injury

Nature of injury	Fatalities	Permanent impairment injuries
2022		
Amputation	1	5
Burn	3	1
Chemical or physical irritation or exposure	0	1
Crushing	12	2
Cut, Puncture, Scrape	2	0
Fracture	1	6
Inhalation, poisoning, intoxication, asphyxiation, drowning	2	0
Loss of consciousness	1	0
Major/multiple system trauma	1	0
Sensory loss/damage (e.g., hearing, sight, taste, smell)	0	1
Shot	1	0
Unspecified	5	0
2023		
Amputation	0	12
Burn	5	0
Chemical or physical irritation or exposure	0	1
Crushing	3	1
Cut, Puncture, Scrape	2	0
Electric shock	1	0
Fracture	0	10
Inhalation, poisoning, intoxication, asphyxiation, drowning	6	0
Major/multiple system trauma	1	0
Organ function/loss or damage	1	2
Paralysis	0	1
Sensory loss/damage (e.g. hearing, sight, taste, smell)	0	1
Sprain, strain	0	2
Unspecified	8	1
2024		
Amputation	0	18
Burn	7	4
Concussion	0	1
Crushing	2	3
Cut, Puncture, Scrape	0	2
Electric shock	2	0
Fracture	0	2
Inhalation, poisoning, intoxication, asphyxiation, drowning	1	0
Major/multiple system trauma	7	2
Organ function / loss or damage	1	1
Sensory loss/damage (e.g. hearing, sight, taste, smell)	0	1
Unspecified	7	1

Table B9: Permanent impairment injuries by category and subcategory

Category	2022	2023	2024	TOTAL
Permanent loss of body parts	6	11	24	41
Amputations with loss of bone inclusive of one or more joints (not able to be reattached)	5	11	20	36
Unspecified	1	0	0	1
Loss of eye(s)	0	0	1	1
Loss of internal organs e.g. kidney, spleen (may also meet criteria 3.4.2)	0	0	2	2
Loss of facial structures, example: nose, ears, leading to disfigurement	0	0	1	1
Permanent reduction of organ function	2	3	2	7
Eye injury with loss of vision	0	1	1	2
Loss of hearing or persistent tinnitus (ear ringing) post injury after exposure to a blast	1	0	0	1
Loss of smell that creates a safety concern or significant personal impact	0	1	0	1
Lung injury, including inhalation, with impaired exertion tolerance/measurable loss of function	1	0	0	1
Spinal injury requiring use of wheelchair	0	1	0	1
Genital or other sexual organ injury leading to loss of function	0	0	1	1
Permanent reduction in skin and/or musculoskeletal function	8	13	9	30
Burns: deep burns with scarring or skin grafts that impair body movement or other function	1	0	5	6
Fracture with ongoing impairment to mobility	6	7	1	14
Other	0	2	1	3
Tendon/ligament tear with ongoing pain and/or inadequate joint range of movement for normal work or general function	0	2	0	2
Traumatic soft tissue injury to shoulder/knee/elbow with ongoing pain and/or inadequate joint range of movement	0	1	1	2
Unspecified	1	1	1	3
Permanent reduction in psychological, social or cognitive function	0	2	2	4
Brain injury with ongoing impact to emotional regulation, speech, memory, or other cognitive dysfunction	0	1	1	2
Other	0	0	1	1
Unspecified	0	1	0	1
Unspecified	0	2	0	2
Unspecified	0	2	0	2

Section 4

Table B10: Fatalities and permanent impairment injuries by region 2022-2024

Region	Fatalities	Permanent impairment injuries
2022		
Africa	2	1
Asia/Australasia	5	1
Europe	0	2
Middle East	11	1
North America	4	9
Russia & Central Asia	3	1
South & Central America	4	1
2023		
Africa		
Asia/Australasia	11	5
Europe	0	7
Middle East	4	7
North America	3	6
Russia & Central Asia	4	1
South & Central America	0	5
2024		
Africa		
Asia/Australasia	2	3
Europe	1	3
Middle East	9	9
North America	3	10
Russia & Central Asia	2	1
South & Central America	2	4

Section 5

Table B11: Fatalities and permanent impairment injuries by work function

Work function	Fatalities*	Permanent impairment injuries
2022		
Exploration	0	0
Drilling	8	7
Production	9	5
Construction	5	2
Unspecified	7	2
2023		
Exploration	0	1
Drilling	2	11
Production	13	12
Construction	5	5
Unspecified	7	2
2024		
Exploration	0	0
Drilling	13	12
Production	13	12
Construction	1	5
Unspecified	0	6

*The work function reported for the fatally injured individuals is shown in Table B11. IOGP Report 2023s – *IOGP Safety performance indicators – 2023 data* and IOGP Report 2024s – *IOGP Safety performance indicators – 2024 data*, state the work function reported for the incident rather than the individual. Therefore, the work function data in these Reports may vary and should not be compared.

Section 6

For reasons of anonymity, each of the companies that has contributed relevant data and is to be included in this analysis has been allocated a unique code letter. These codes change every year in line with FPIR performance.

Table B12: Fatality and permanent impairment rate (FPIR) and permanent impairment rate (PIR) by company

2022				2023				2024			
Company code	FPIR	PIR	FAR	Company code	FPIR	PIR	FAR	Company code	FPIR	PIR	FAR
A	18.48	18.48	0	*A	93.56	0	93.56	*A	57.73	0	57.73
*B	12.75	6.37	6.37	B	37.69	37.69	0	*B	33.62	0	33.62
*C	12.18	0	12.18	C	10.97	10.97	0	*C	15.08	0	15.08
D	7.53	7.53	0	*D	9.56	4.78	4.78	D	12.01	12.01	0
*E	5.69	0	5.69	E	8.53	8.53	0	E	6.76	6.76	0
F	5.23	5.23	0	F	5.71	5.71	0	F	4.62	4.62	0
*G	3.15	2.1	1.05	*G	5.62	4.5	1.12	G	4.16	4.16	0
*H	2.32	0.77	1.55	*H	2.75	1.38	1.38	H	3.72	3.72	0
*I	2.04	0	2.04	*I	2.66	1.06	1.59	*I	3.64	0	3.64
*J	1.96	0	1.96	*J	2.01	0.67	1.34	J	3.13	3.13	0
K	1.96	1.96	0	K	1.97	1.97	0	*K	2.91	1.94	0.97
Overall	1.93	0.69	1.24	Overall	1.78	0.95	0.83	*L	2.31	1.54	0.77
*L	1.85	0.62	1.24	L	1.74	1.74	0	M	1.95	1.95	0
*M	1.43	0	1.43	M	1.52	1.52	0	*N	1.84	0.92	0.92
*N	1.07	0.54	0.54	*N	1.33	0	1.33	O	1.83	1.83	0
*O	0.6	0	0.6	O	1.21	1.21	0	Overall	1.81	1.02	0.79
P	0	0	0	*P	1.06	0	1.06	*P	1.80	1.20	0.60
Q	0	0	0	Q	0.96	0.96	0	*Q	1.72	1.29	0.43
R	0	0	0	*R	0.93	0	0.93	*R	1.55	1.16	0.39
S	0	0	0	*S	0.86	0.43	0.43	*S	1.54	0	1.54
T	0	0	0	*T	0.81	0	0.81	T	1.38	1.38	0
U	0	0	0	U	0.79	0.79	0	*U	1.08	0.54	0.54
V	0	0	0	V	0	0	0	*V	0.91	0.45	0.45
W	0	0	0	W	0	0	0	*W	0.70	0.23	0.47
X	0	0	0	X	0	0	0	*X	0.46	0	0.46
Y	0	0	0	Y	0	0	0	Y	0	0	0
Z	0	0	0	Z	0	0	0	Z	0	0	0
AA	0	0	0	AA	0	0	0	AA	0	0	0

2022				2023				2024			
Company code	FPIR	PIR	FAR	Company code	FPIR	PIR	FAR	Company code	FPIR	PIR	FAR
BB	0	0	0	BB	0	0	0	BB	0	0	0
CC	0	0	0	CC	0	0	0	CC	0	0	0
DD	0	0	0	DD	0	0	0	DD	0	0	0
EE	0	0	0	EE	0	0	0	EE	0	0	0
FF	0	0	0	FF	0	0	0	FF	0	0	0
GG	0	0	0	GG	0	0	0	GG	0	0	0
HH	0	0	0	HH	0	0	0	HH	0	0	0
II	0	0	0	II	0	0	0	II	0	0	0
				JJ	0	0	0	JJ	0	0	0
				KK	0	0	0	KK	0	0	0
				LL	0	0	0	LL	0	0	0
				MM	0	0	0	MM	0	0	0
				NN	0	0	0	NN	0	0	0
				OO	0	0	0	OO	0	0	0
				PP	0	0	0	PP	0	0	0
				QQ	0	0	0	QQ	0	0	0
				RR	0	0	0	RR	0	0	0
				SS	0	0	0	SS	0	0	0
				TT	0	0	0	TT	0	0	0
				UU	0	0	0	UU	0	0	0
				VV	0	0	0	VV	0	0	0
				WW	0	0	0				
				XX	0	0	0				
				YY	0	0	0				
				ZZ	0	0	0				
				AAA	0	0	0				
* indicates fatality in reporting year											

References

American Medical Association (AMA) Guides to the Evaluation of Permanent Impairment, 6th edition.

FPI definition, examples, and associated Frequently Asked Questions (FAQ) are available at <https://iogp.org/fpi>

IOGP Report 2022s - *Safety Performance Indicators - 2022 data*

IOGP Report 2023s - *Safety Performance Indicators - 2023 data*

IOGP Report 2024s - *Safety Performance Indicators - 2024 data*

IOGP Report 459 - *Life-Saving Rules*

IOGP Life-Saving Rules and supporting materials:
<https://www.iogp.org/workstreams/safety/safety/life-savingrules/>

Peuscher, W., Groeneweg, J., A Big Oil Company's Approach to Significantly Reduce Fatal Incidents, Paper presented at the International Conference on Health, Safety and Environment in Oil and Gas Exploration and Production, 11–13 September 2012, SPE-157465-MS, <https://onepetro.org/SPEHSE/proceedings-abstract/12HSE/12HSE/SPE-157465-MS/158675>

Walker, K., et al., "Focusing Industry Learning on Prevention of Fatalities and Life-changing Injuries", SPE-220314-MS, *SPE International Health, Safety, Environment and Sustainability Conference and Exhibition*, Abu Dhabi, UAE, September 2024, (see <https://onepetro.org/SPEHSE/proceedings/24HSE/24HSE/D011S008R003/555541>)

World Health Organization (WHO) [International Classification of Functioning, Disability and Health \[ICF\]](#).

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